Coilem A/B/C/D benchmark

YELLOW

Evaluation protocol 1.3.0: winding-phase-a-v1. Measured protocol remains 1.1.0. Post hoc angle-contract reconciliation; historical FAIL evaluations are retained in JSON. Scientific review and release qualification remain separate.

Harness 1.3.0; protocol 1.1.0.

Candidate commit: 738d9516e8500c5996c772bd96470472cbb9a0f9

Reference: FROZEN_REFERENCE; commit 738d9516e8500c5996c772bd96470472cbb9a0f9.

Orange solid: Coilem; blue dashed: saved FEMM. No reference solves executed.

Phase	Recorded outcomes
A – SPM	10 PASS
B – IPM	11 PASS, 1 YELLOW
C – Distributed Winding	4 PASS, 1 Planned coverage (not yet supported)
D – BLDC	3 PASS

Phase summary: green for all PASS; yellow for mixed or missing coverage; red for more than 50% FAIL.

Phase A – SPM results

Fixture	Outcome
spm_4p12s_simple__baseline	PASS
spm_4p12s_simple__high_current	PASS
spm_4p12s_simple__tight_airgap	PASS
spm_4p12s_simple_noload__baseline	PASS
spm_4p12s_simple_noload__high_current	PASS
spm_4p12s_simple_noload__tight_airgap	PASS
spm_8p12s_concentrated__baseline	PASS
spm_8p12s_concentrated__high_current	PASS
spm_8p12s_concentrated__tight_airgap	PASS
spm_8p12s_reference_angle30__baseline	PASS

Phase B – IPM results

Fixture	Outcome
ipm_flat_12s10p_fscw_contract	PASS
ipm_flat_24s16p_industrial_contract	PASS

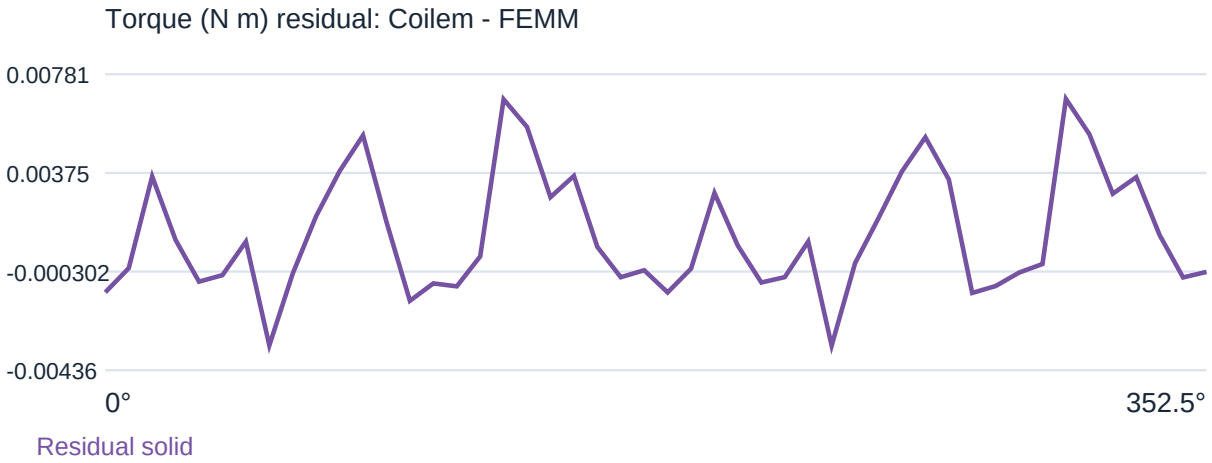
Fixture	Outcome
ipm_flat_4p12s_micro_contract	PASS
ipm_flat_8p12s_fsae_angled_pocket_contract	PASS
ipm_flat_8p12s_fsae_contract	PASS
ipm_flat_8p12s_fsae_thin_bridge_contract	PASS
ipm_vshape_8p12s_baseline_contract	YELLOW
ipm_vshape_8p12s_deep_apex_high_output_v1	PASS
ipm_vshape_8p12s_narrow_angle_contract	PASS
ipm_vshape_8p12s_pocket_offset_contract	PASS
ipm_vshape_8p12s_thin_bridge_contract	PASS
ipm_vshape_8p12s_wide_angle_high_output_v1	PASS

Phase C – Distributed Winding results

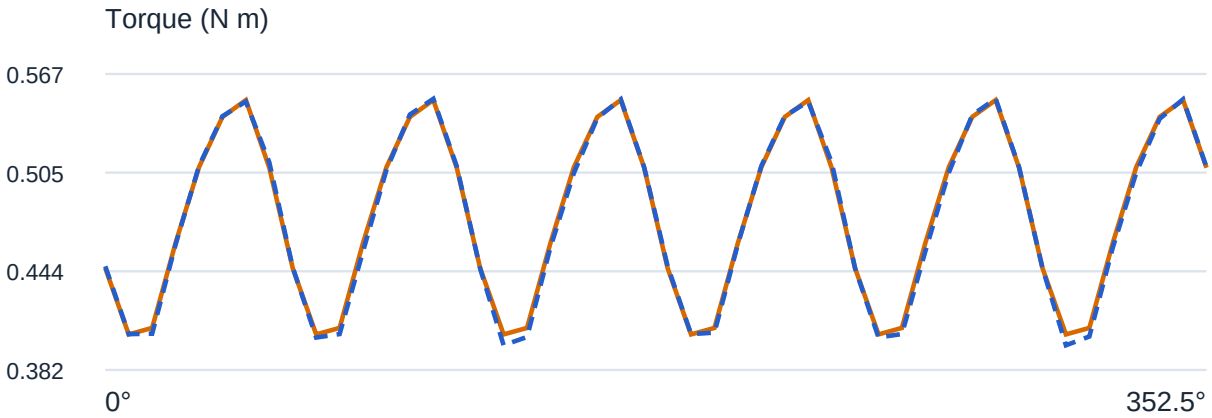
Fixture	Outcome
ornl_prius_2004_ipm	Planned coverage (not yet supported)
spm_12p36s_distributed_femm_validation	PASS
spm_8p24s_distributed_femm_validation	PASS
spm_8p48s_chorded_5_6	PASS
spm_8p48s_distributed_femm_validation	PASS

Phase A – spm_4p12s_simple__baseline

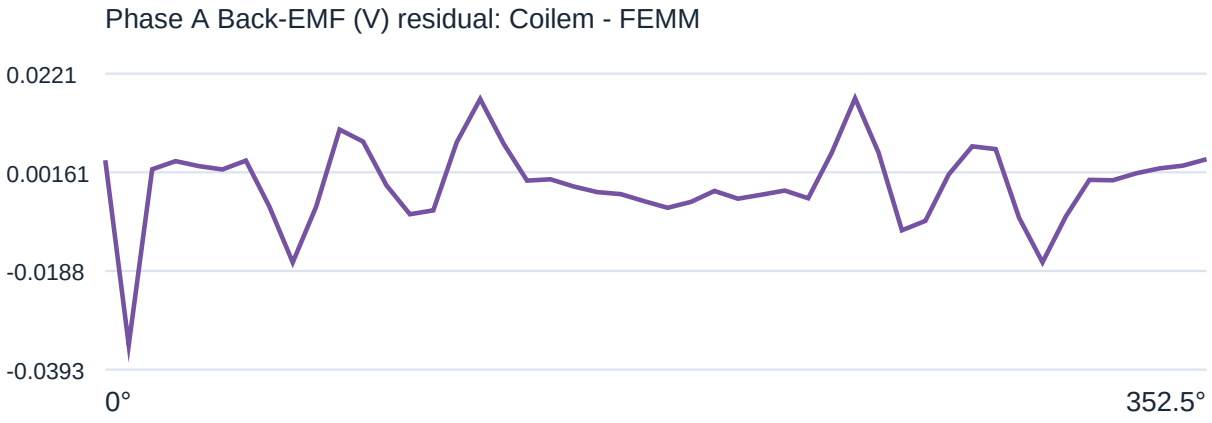
Frozen native fixture JSON



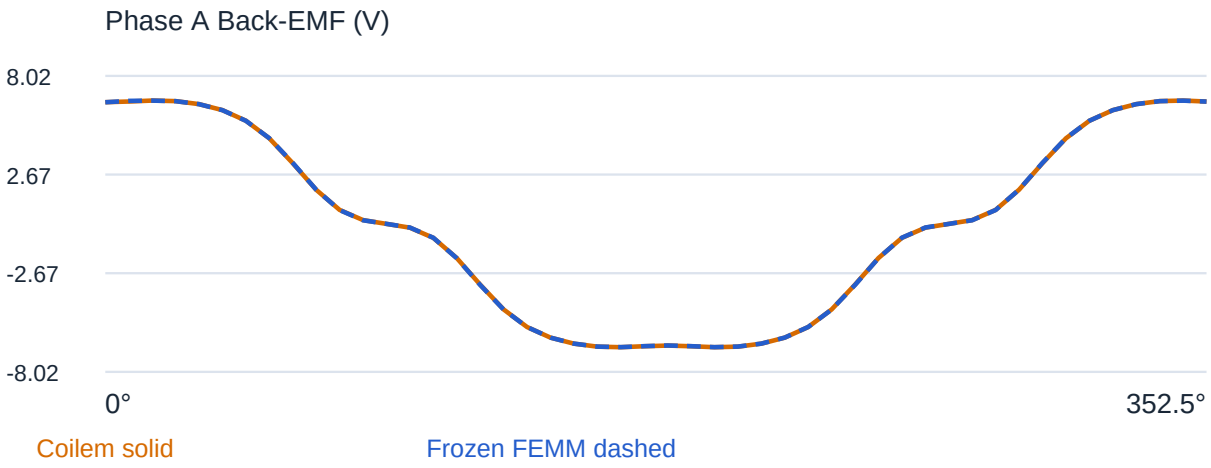
Residual solid



Coilem solid Frozen FEMM dashed



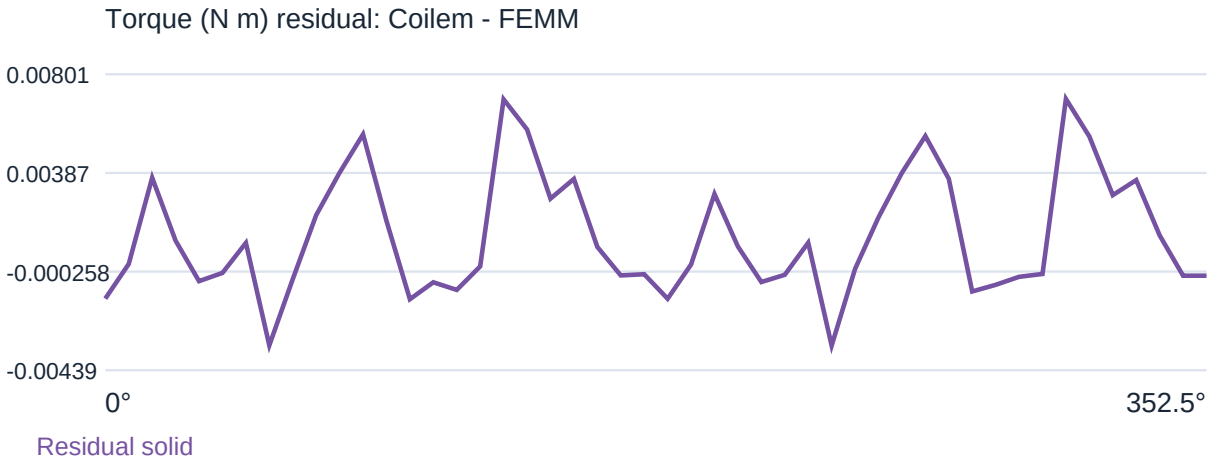
Residual solid



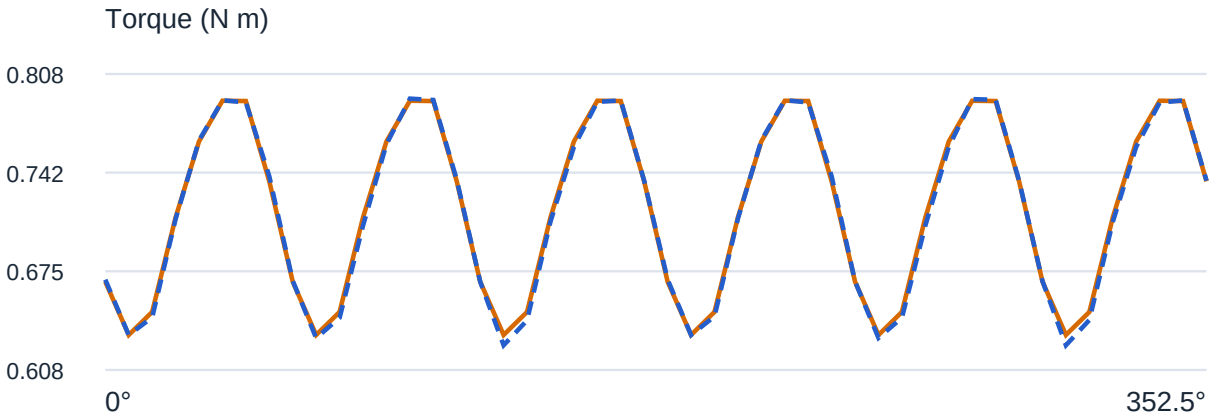
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase A – spm_4p12s_simple__high_current

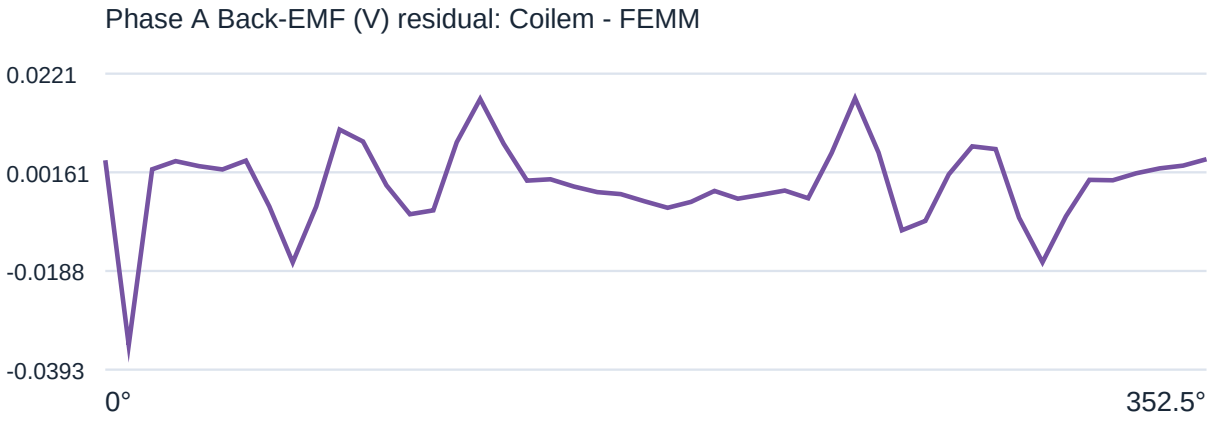
Frozen native fixture JSON



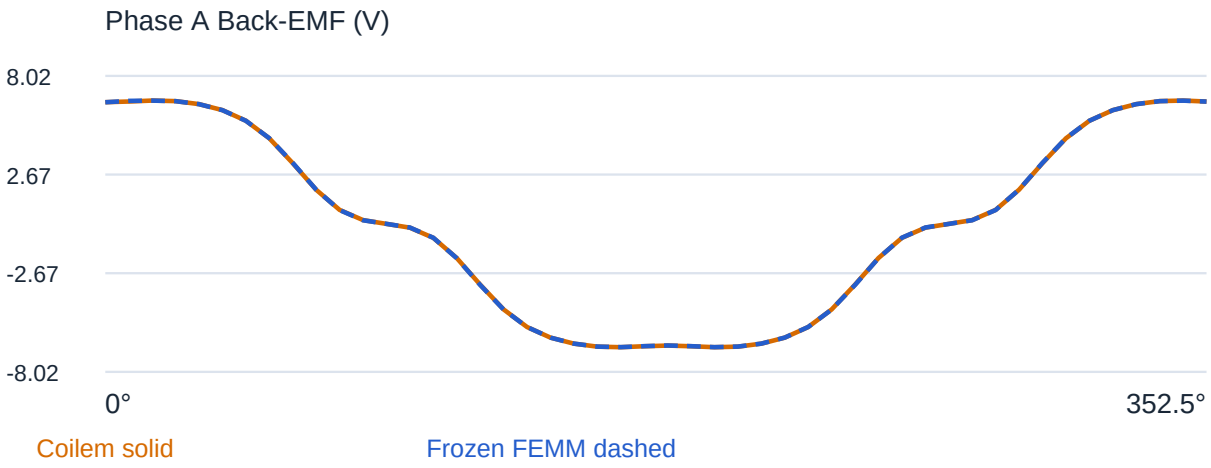
Residual solid



Coilem solid Frozen FEMM dashed



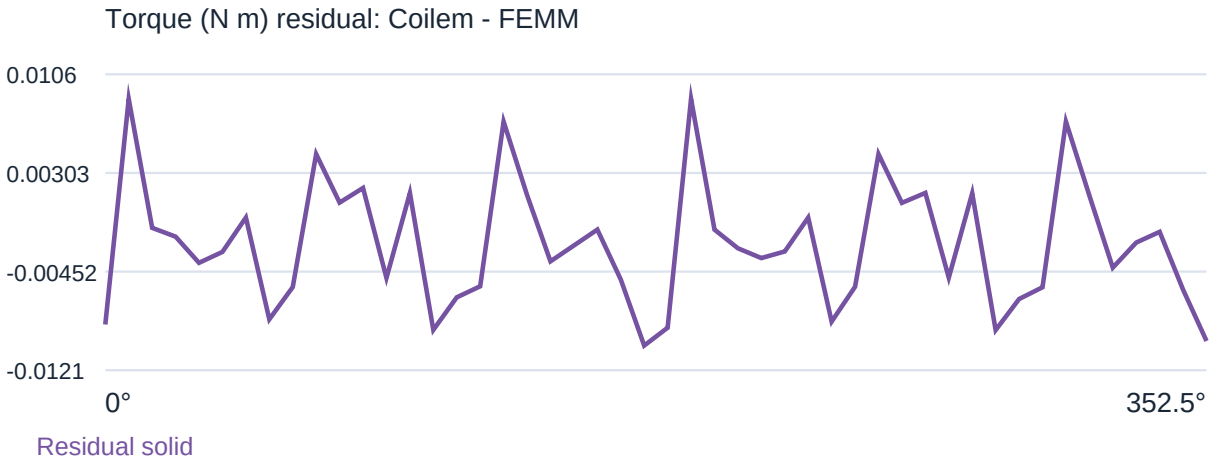
Residual solid



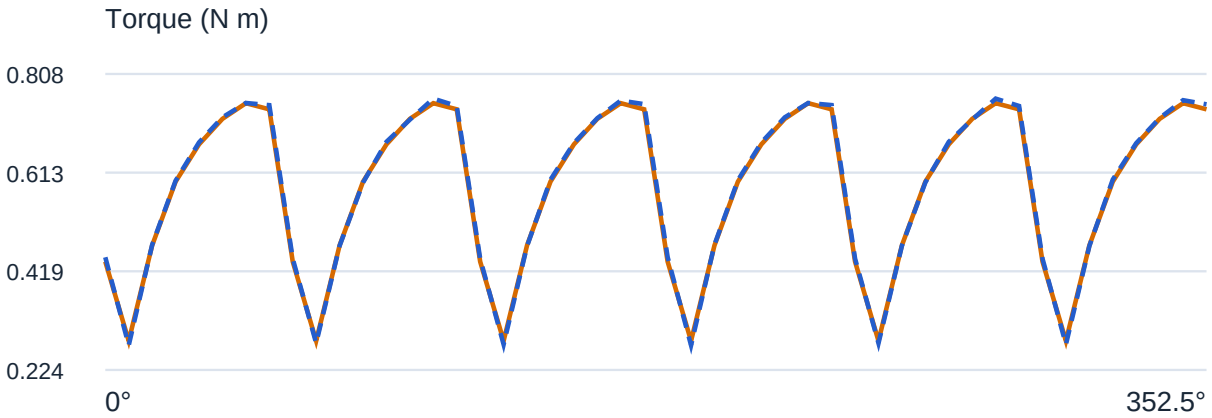
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase A – spm_4p12s_simple__tight_airgap

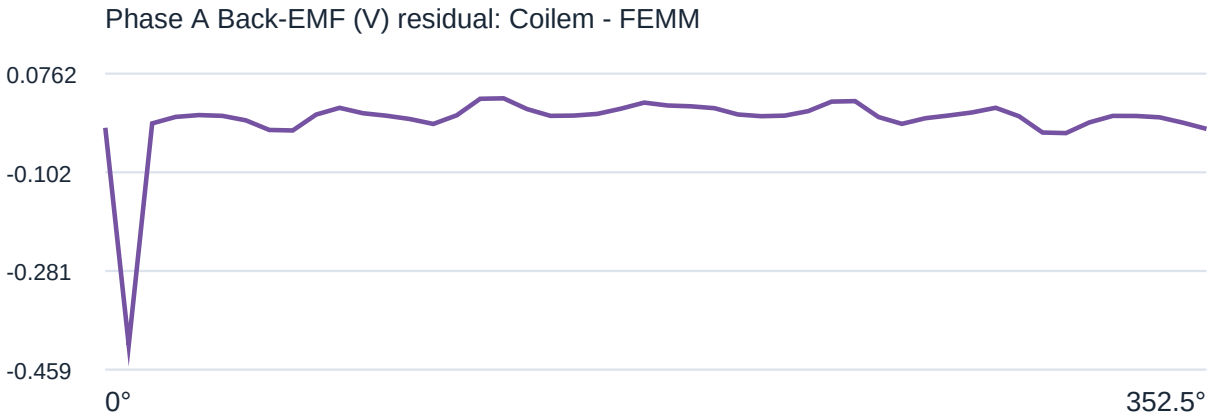
Frozen native fixture JSON



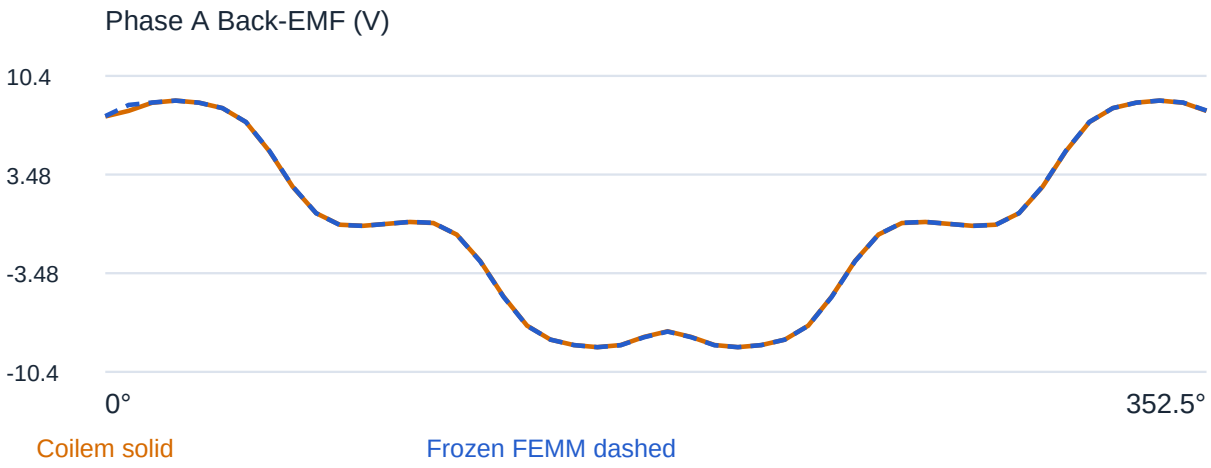
Residual solid



Coilem solid Frozen FEMM dashed



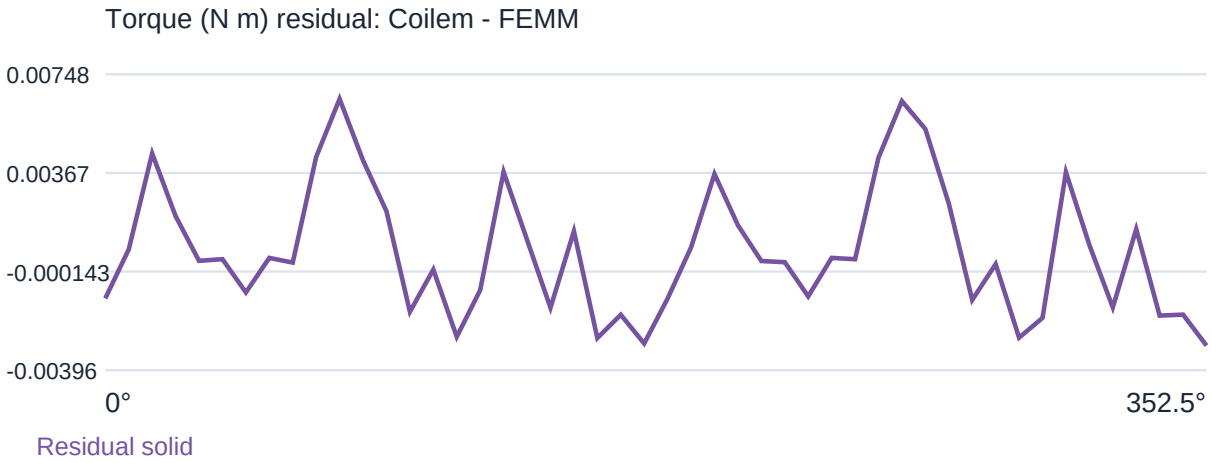
Residual solid



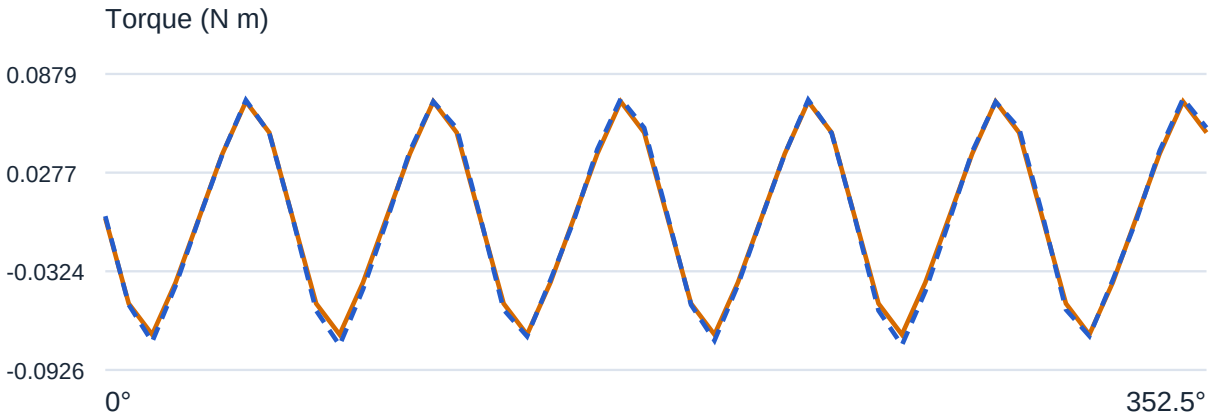
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase A – spm_4p12s_simple_noload__baseline

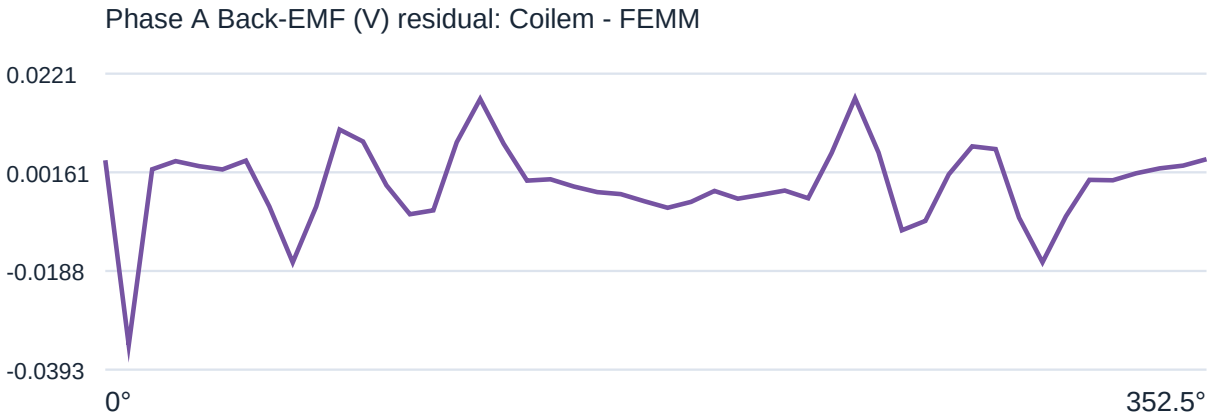
Frozen native fixture JSON



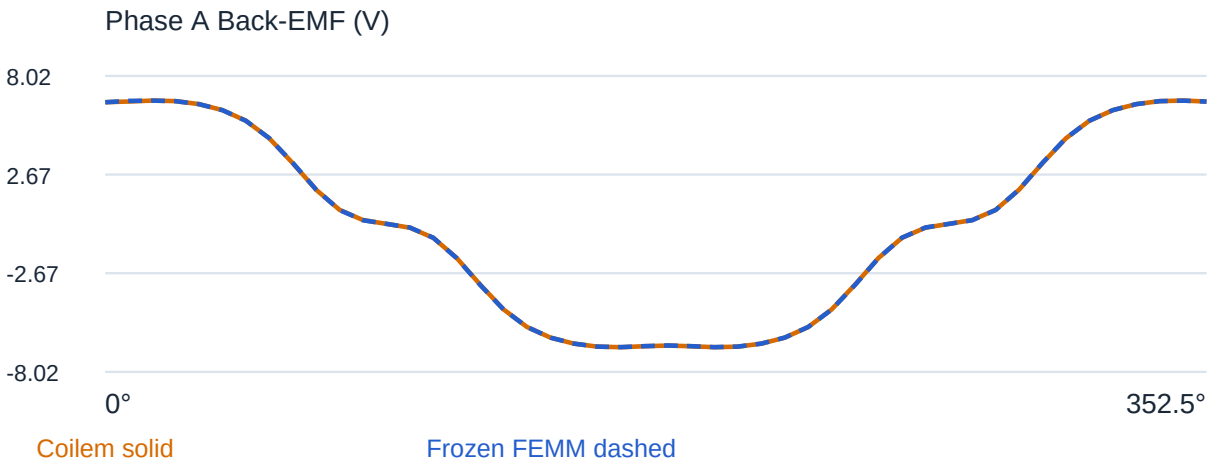
Residual solid



Coilem solid Frozen FEMM dashed



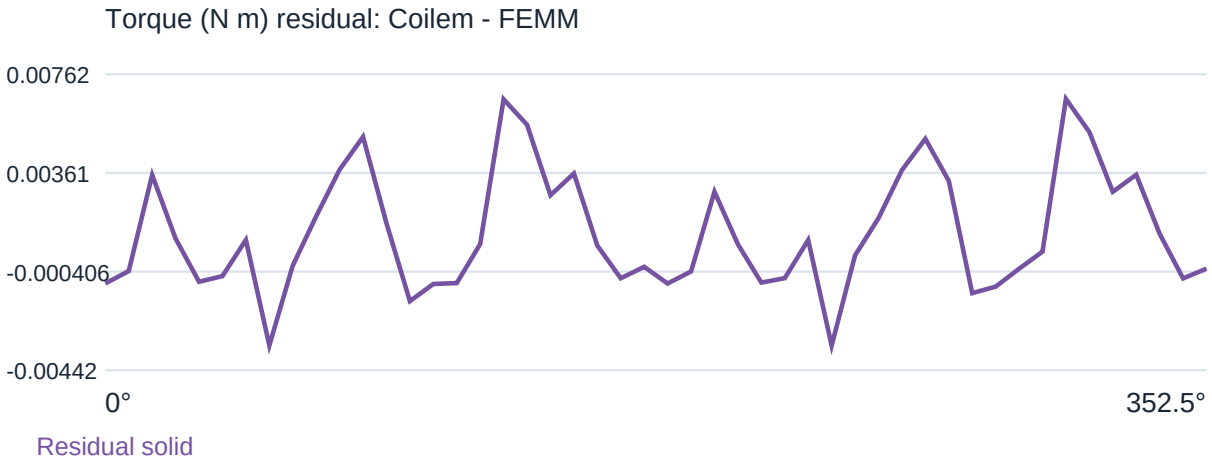
Residual solid



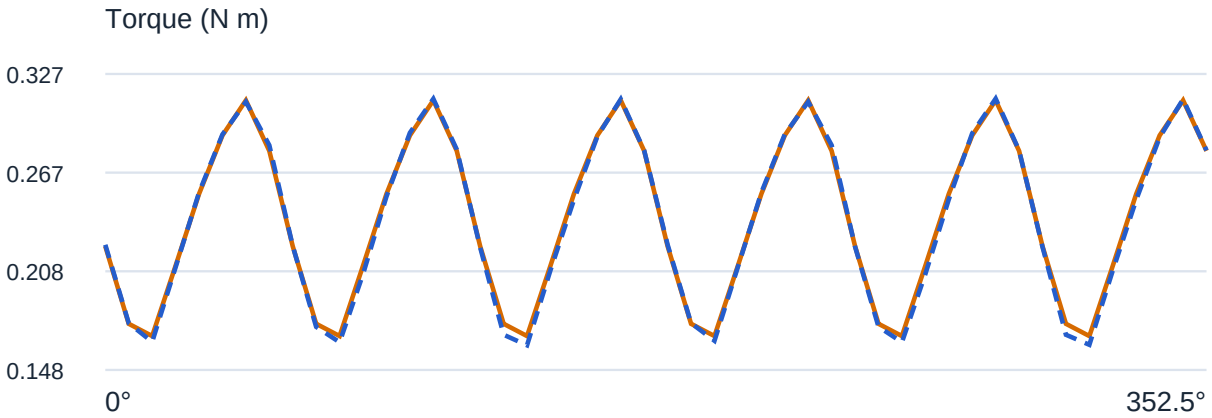
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase A – spm_4p12s_simple_noload__high_current

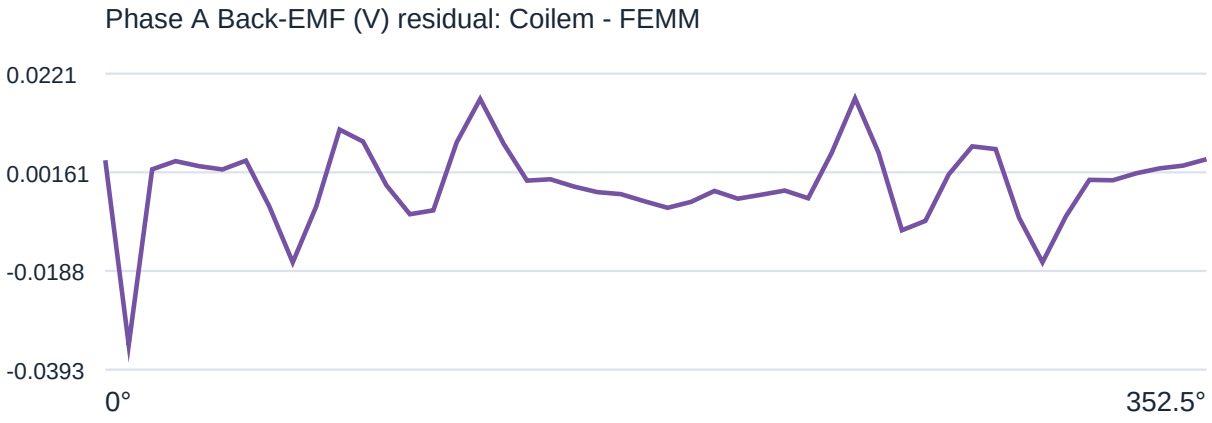
Frozen native fixture JSON



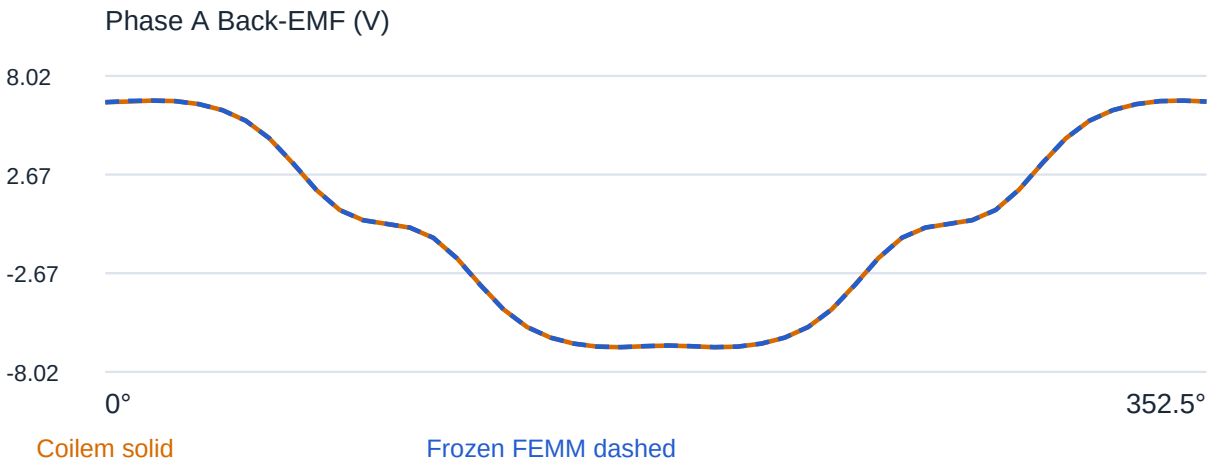
Residual solid



Coilem solid Frozen FEMM dashed



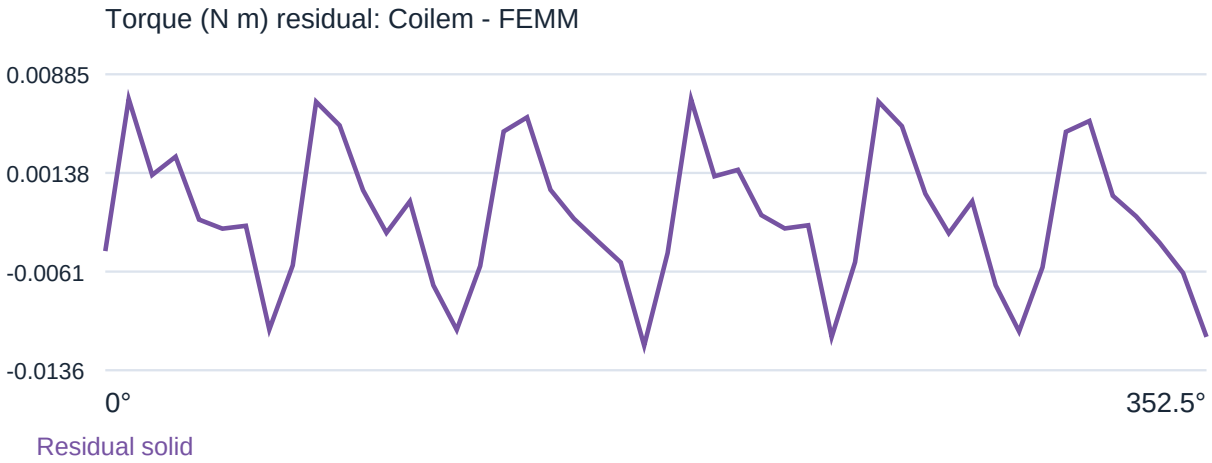
Residual solid



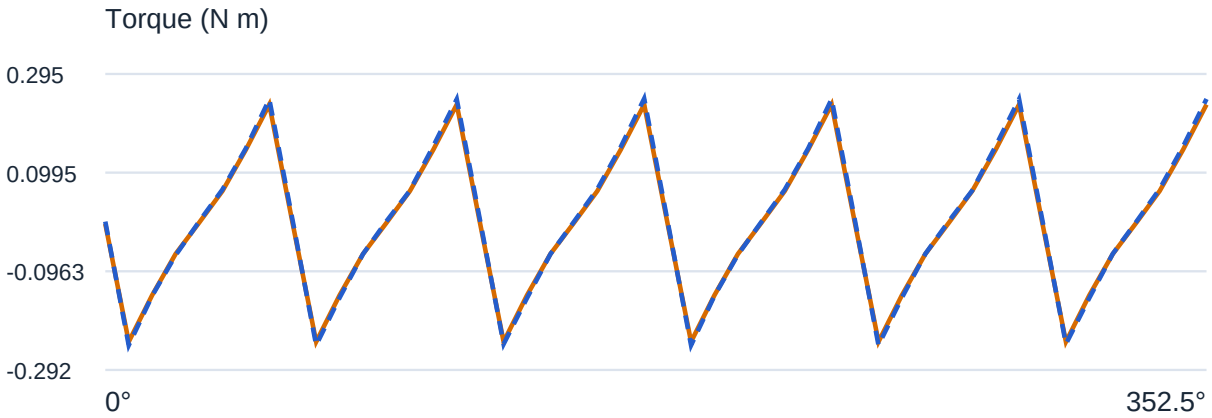
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase A – spm_4p12s_simple_noload__tight_airgap

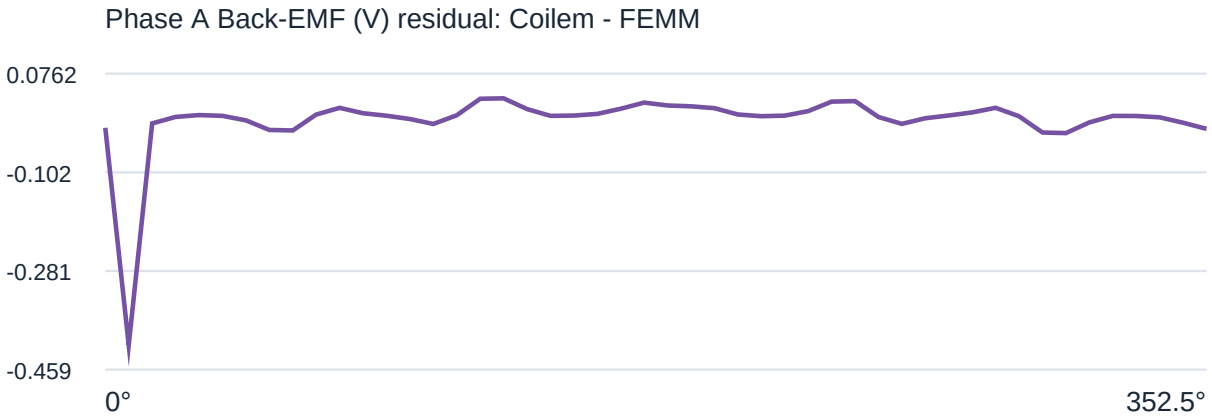
Frozen native fixture JSON



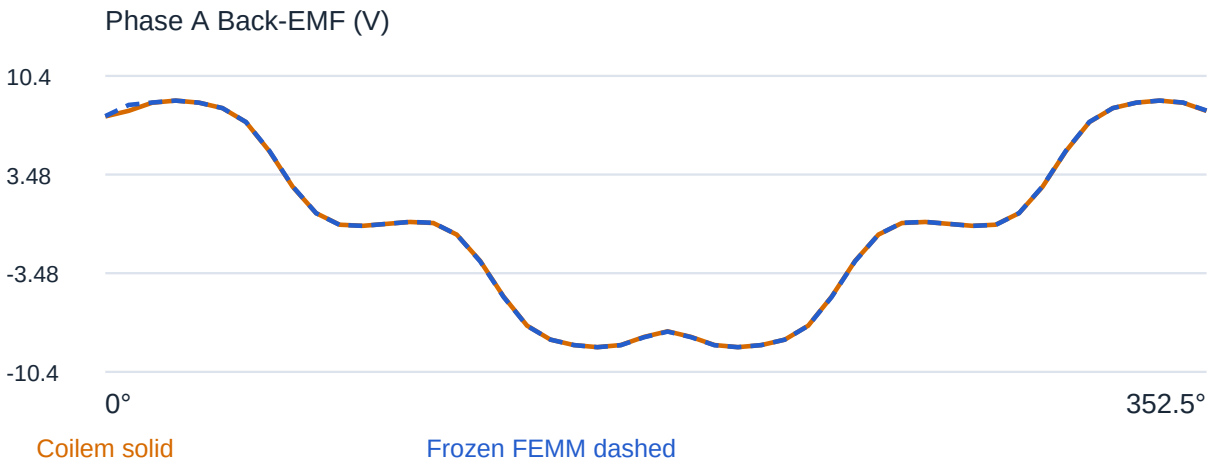
Residual solid



Coilem solid Frozen FEMM dashed



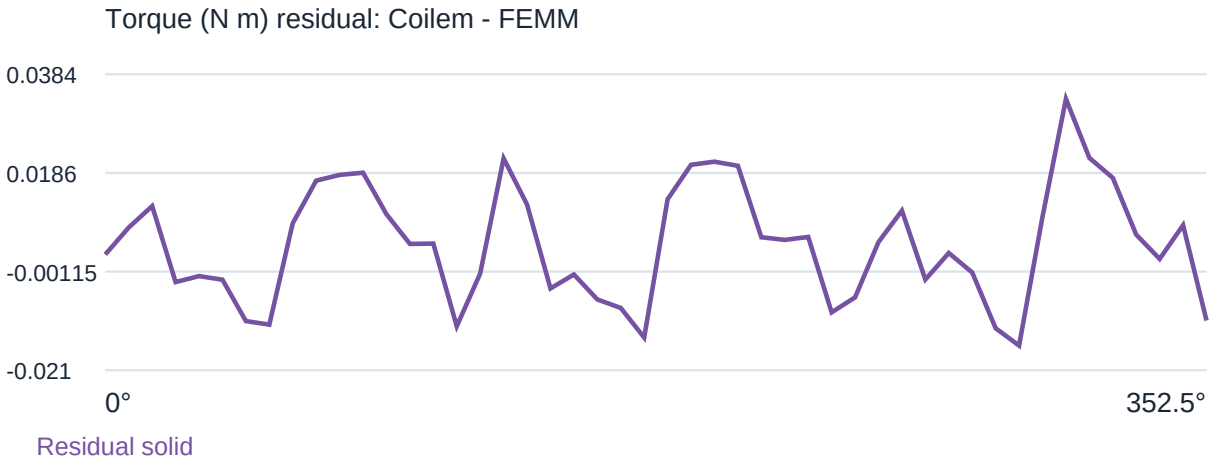
Residual solid



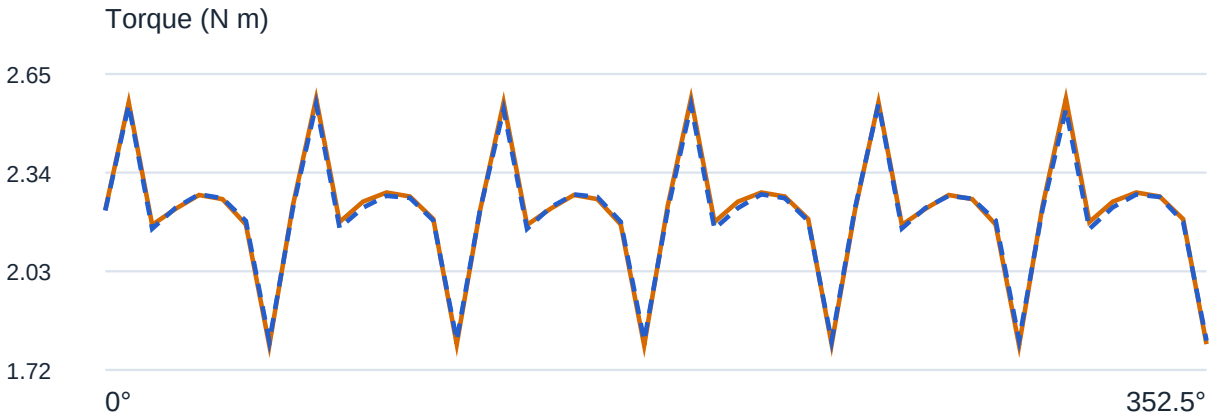
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase A – spm_8p12s_concentrated__baseline

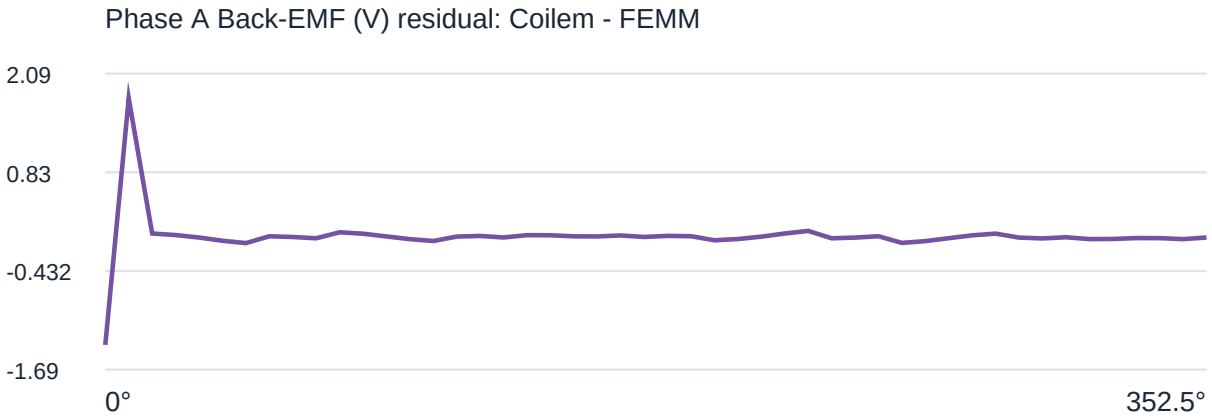
Frozen native fixture JSON



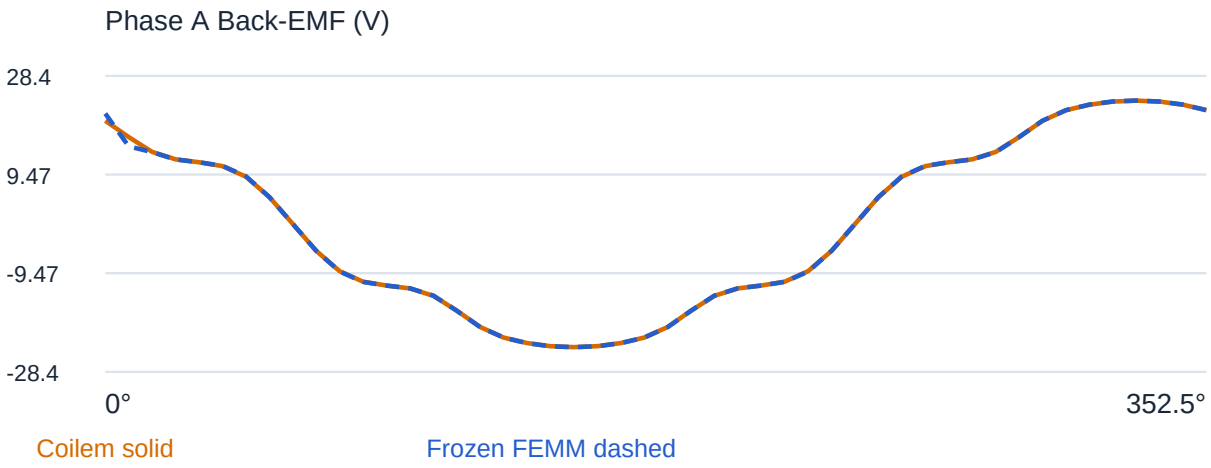
Residual solid



Coilem solid Frozen FEMM dashed



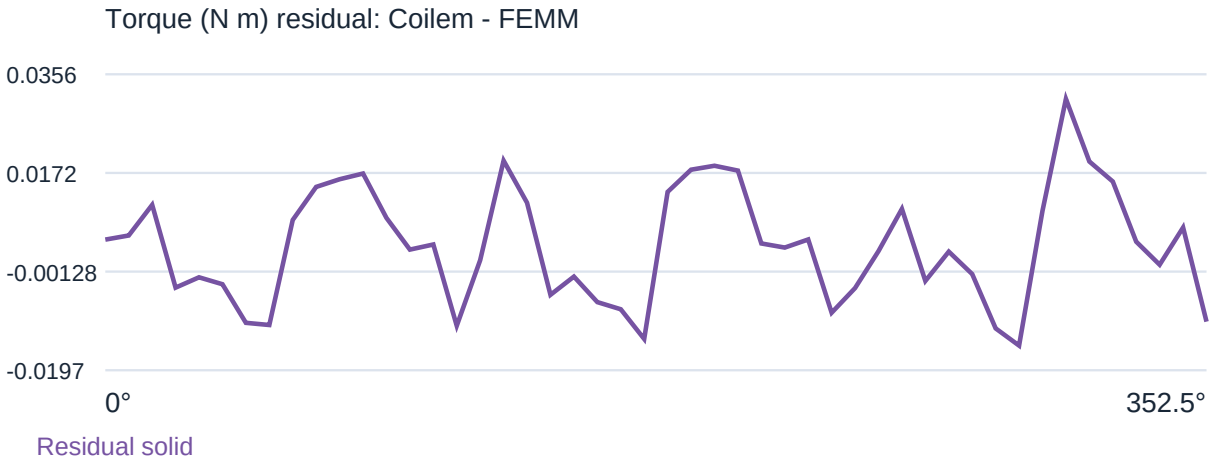
Residual solid



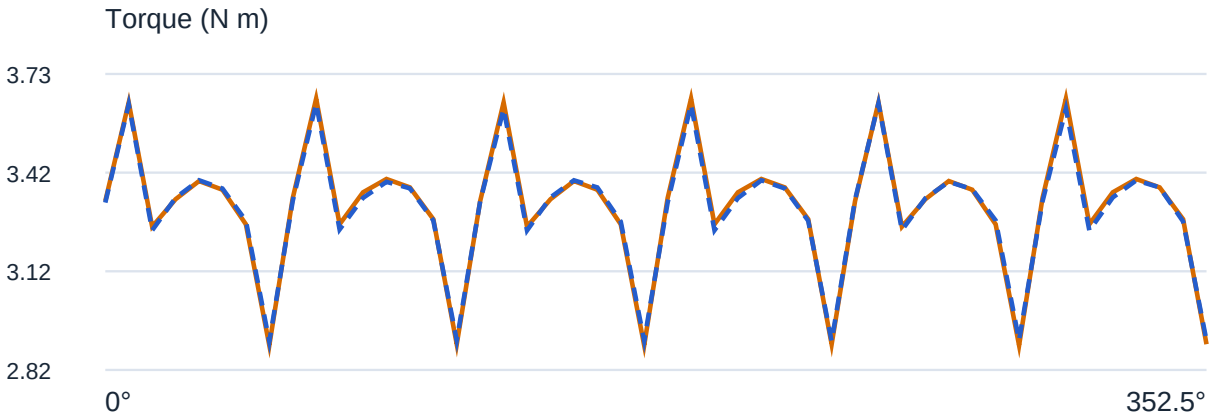
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase A – spm_8p12s_concentrated__high_current

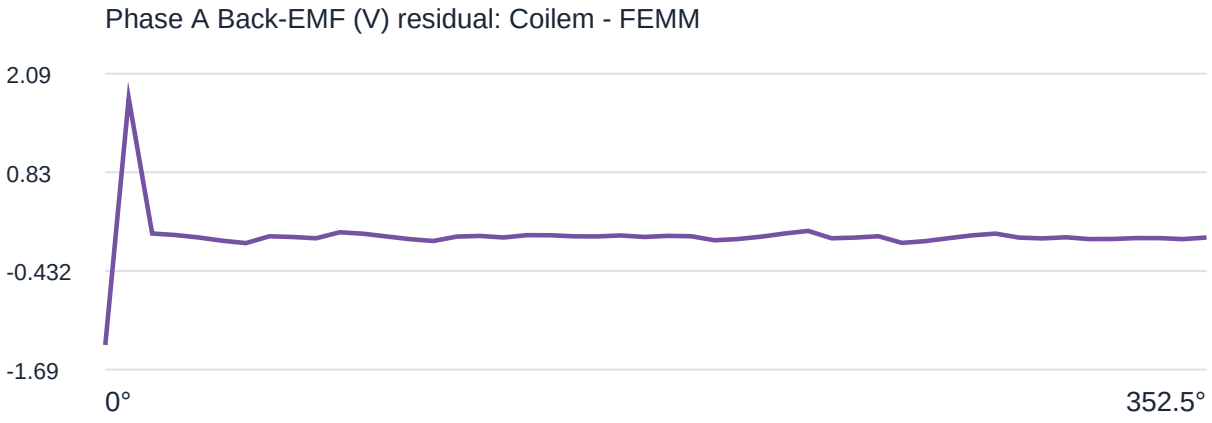
Frozen native fixture JSON



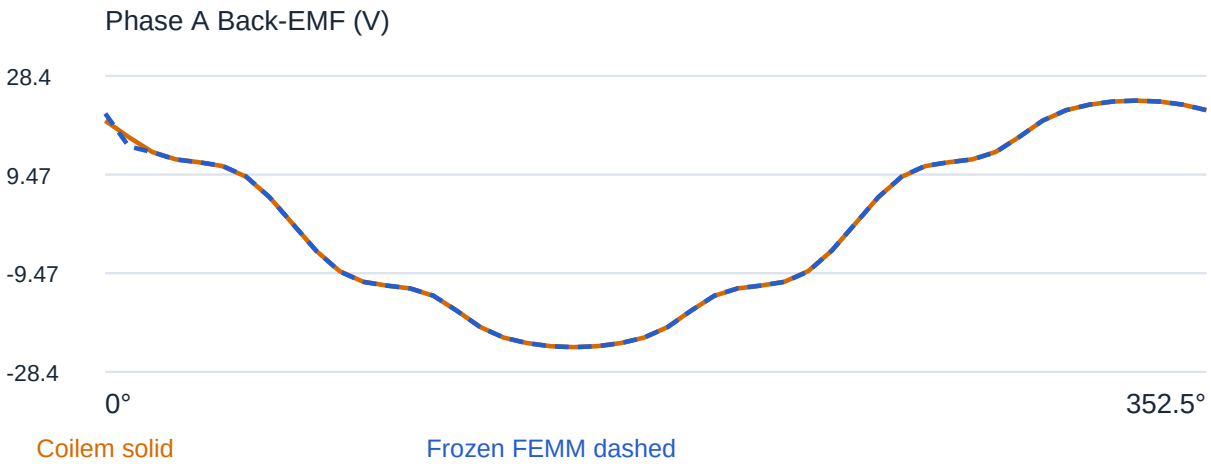
Residual solid



Coilem solid Frozen FEMM dashed



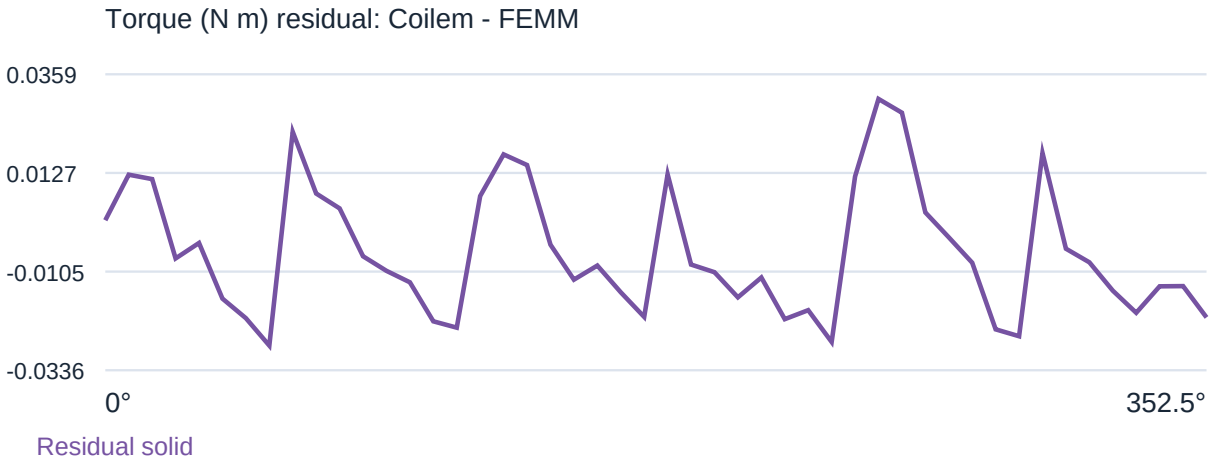
Residual solid



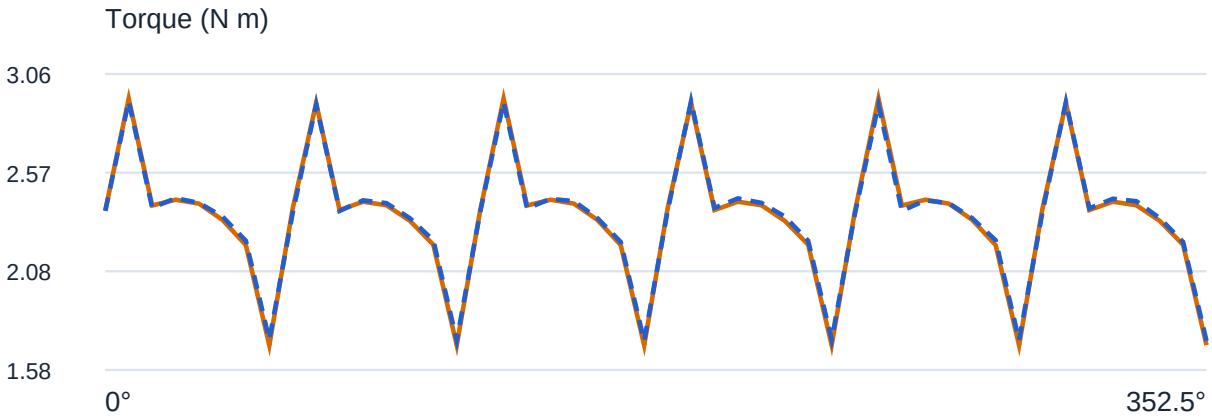
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase A – spm_8p12s_concentrated__tight_airgap

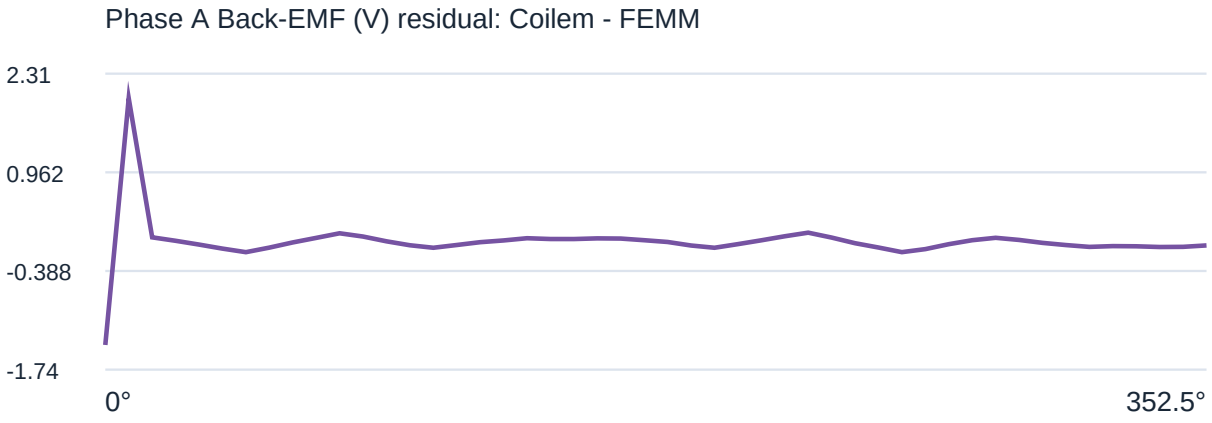
Frozen native fixture JSON



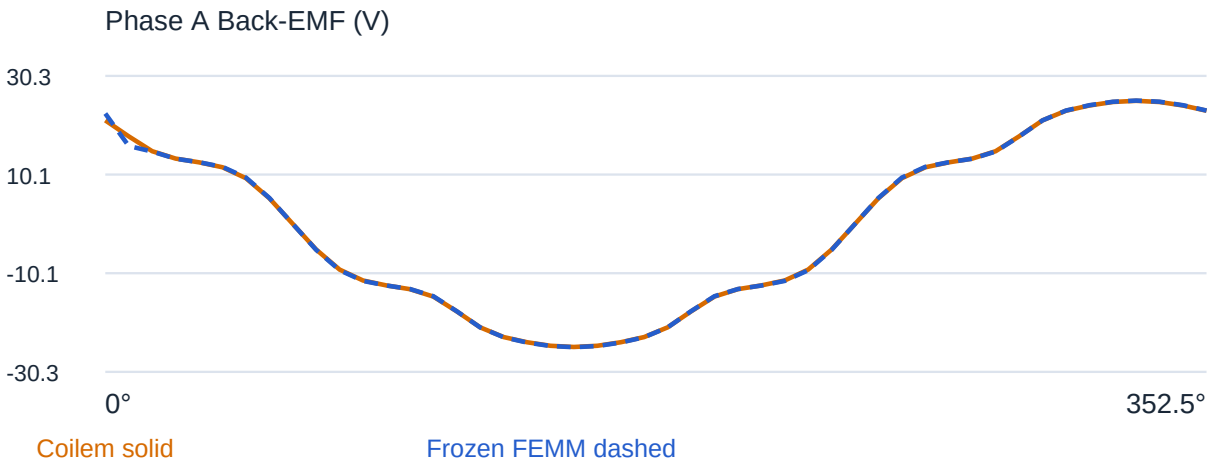
Residual solid



Coilem solid Frozen FEMM dashed



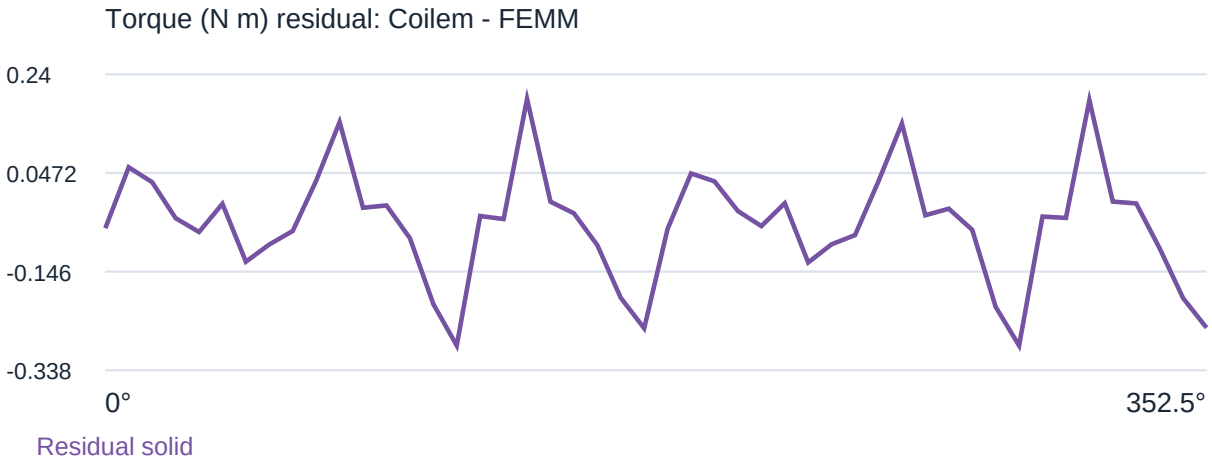
Residual solid



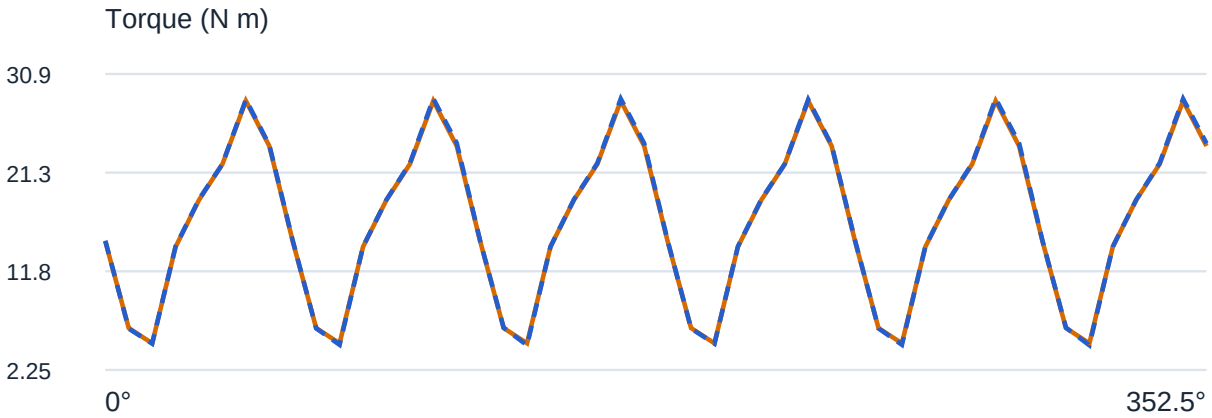
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase A – spm_8p12s_reference_angle30__baseline

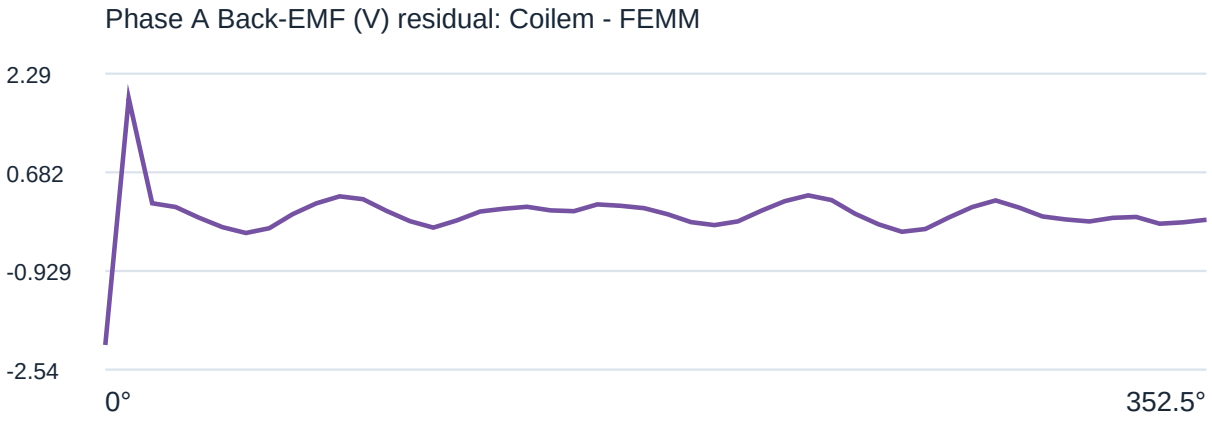
Frozen native fixture JSON



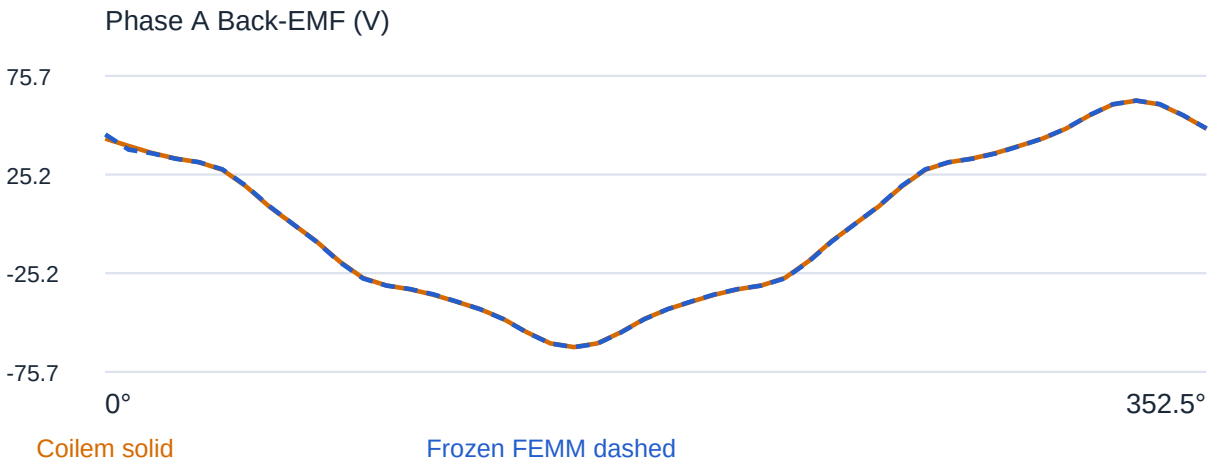
Residual solid



Coilem solid Frozen FEMM dashed



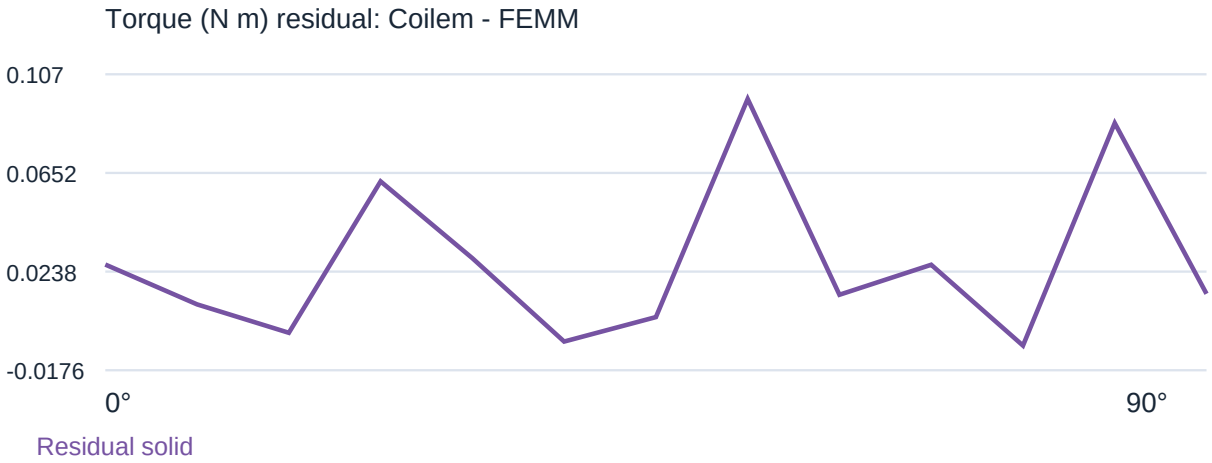
Residual solid



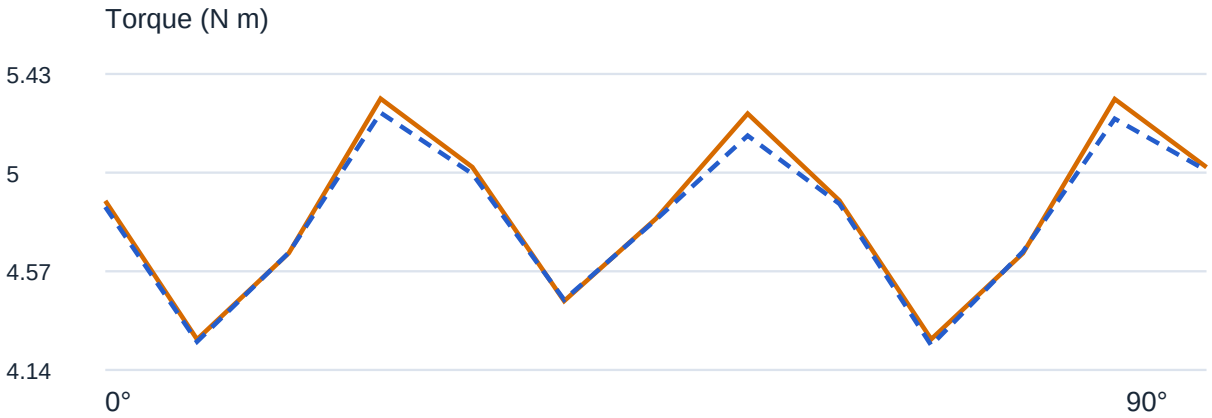
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_flat_12s10p_fscw_contract

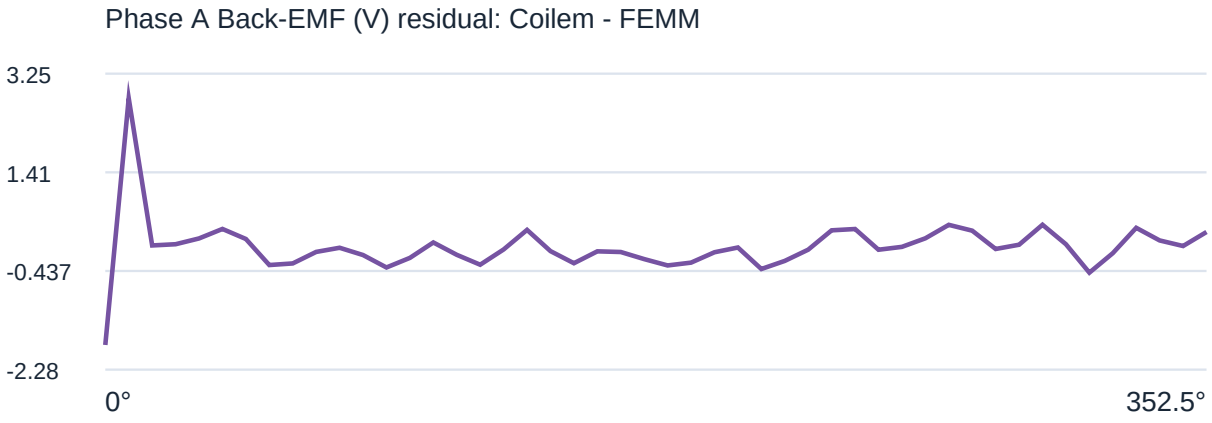
Frozen native fixture JSON



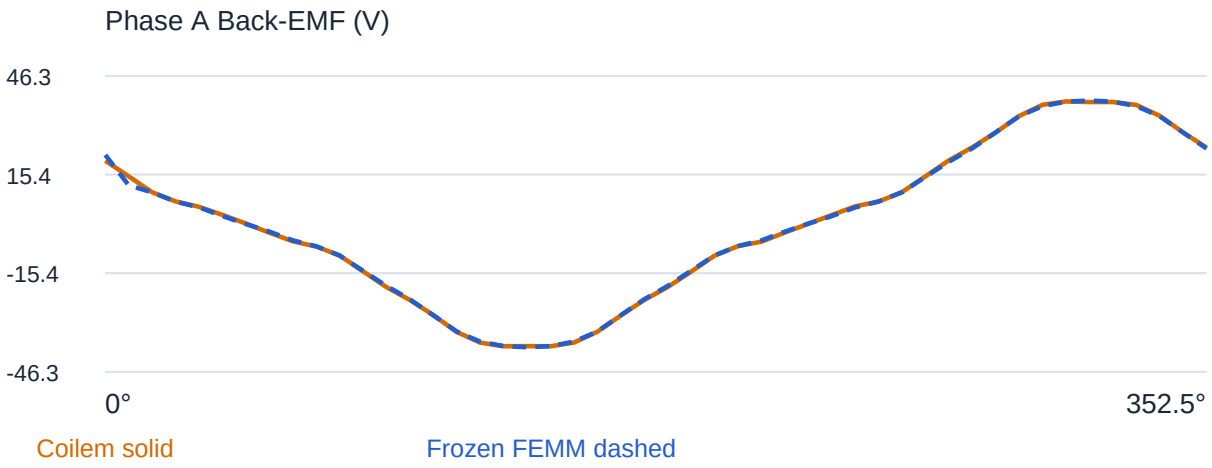
Residual solid



Coilem solid Frozen FEMM dashed



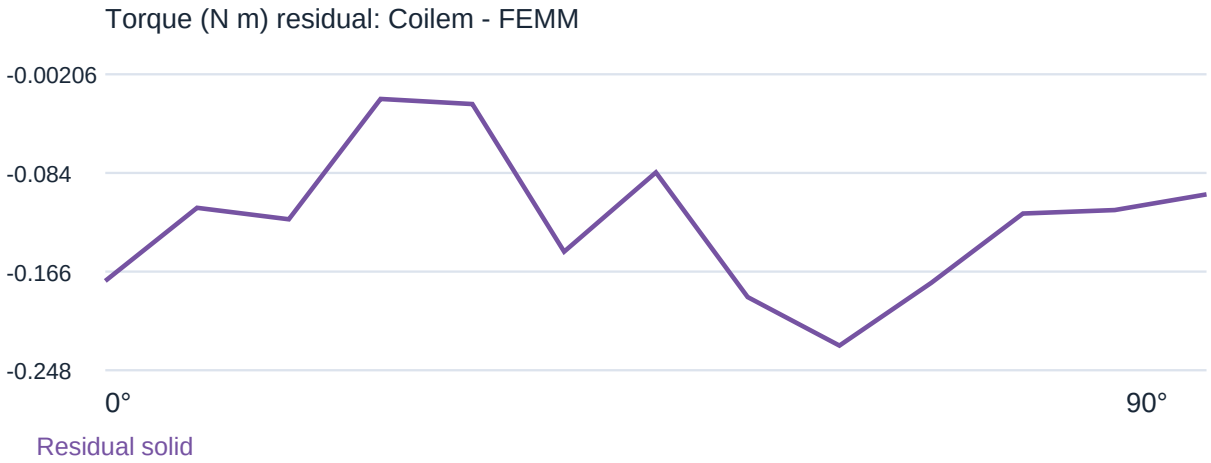
Residual solid



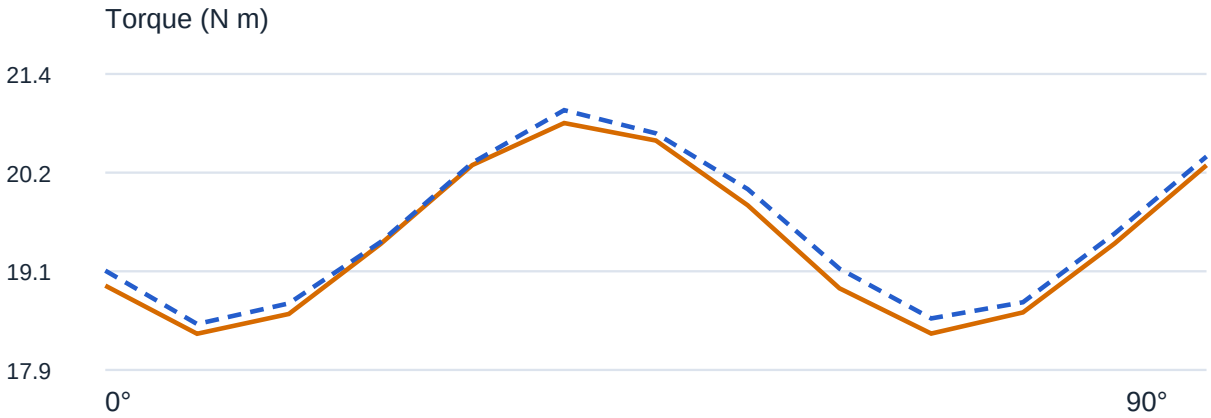
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_flat_24s16p_industrial_contract

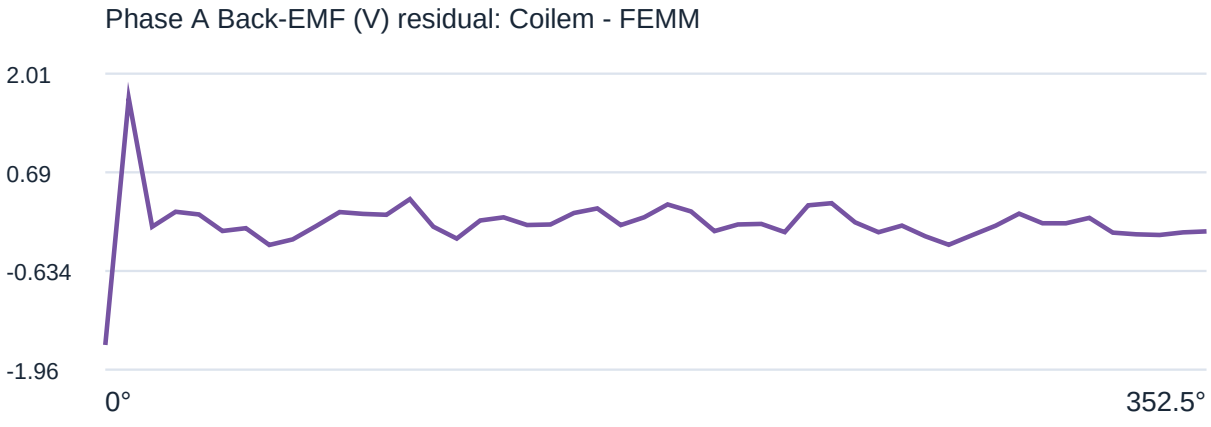
Frozen native fixture JSON



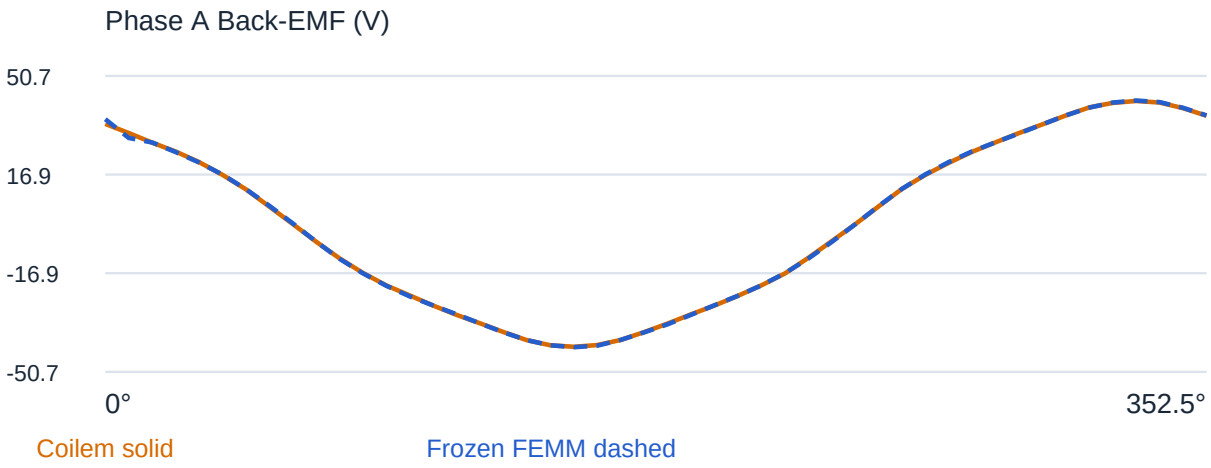
Residual solid



Coilem solid Frozen FEMM dashed



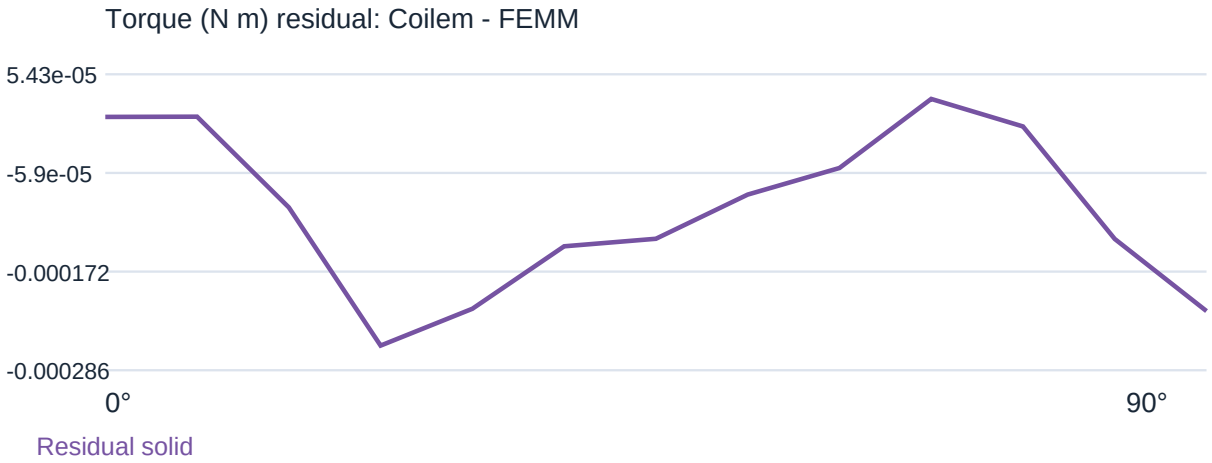
Residual solid



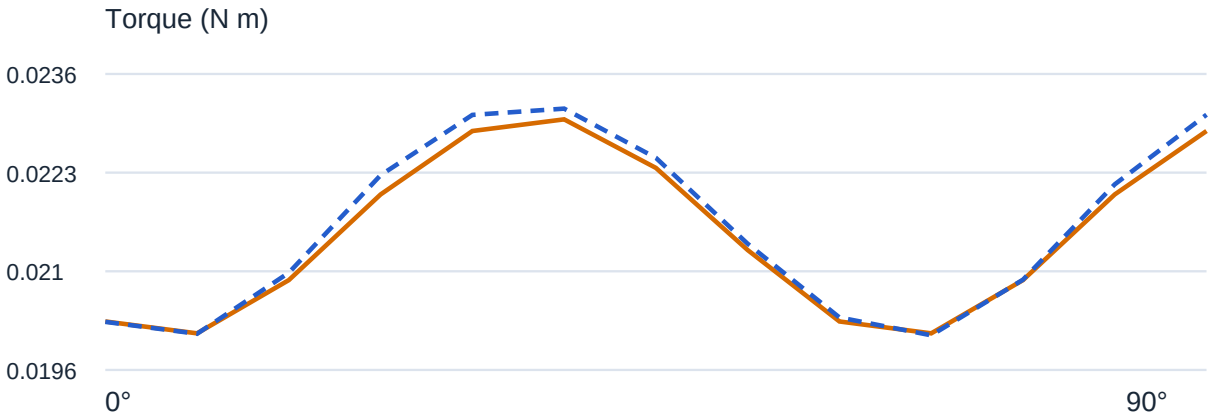
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_flat_4p12s_micro_contract

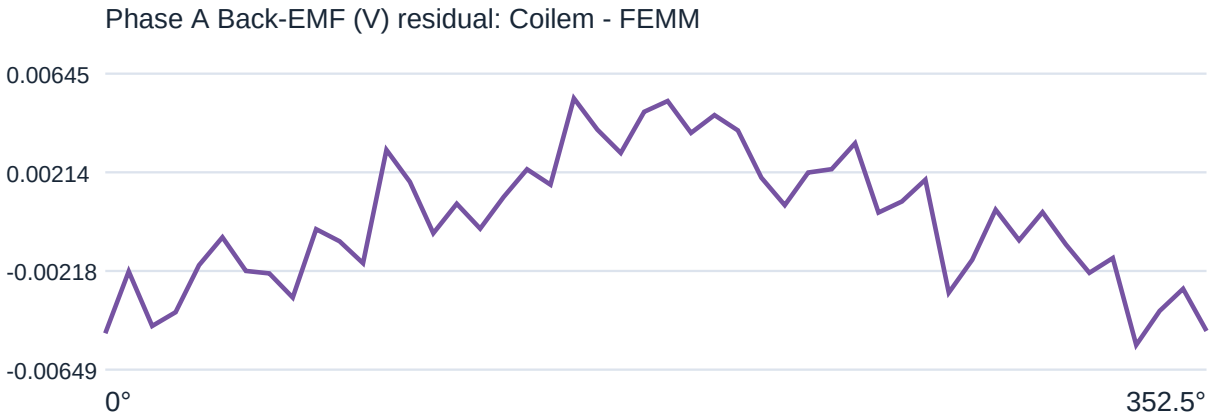
Frozen native fixture JSON



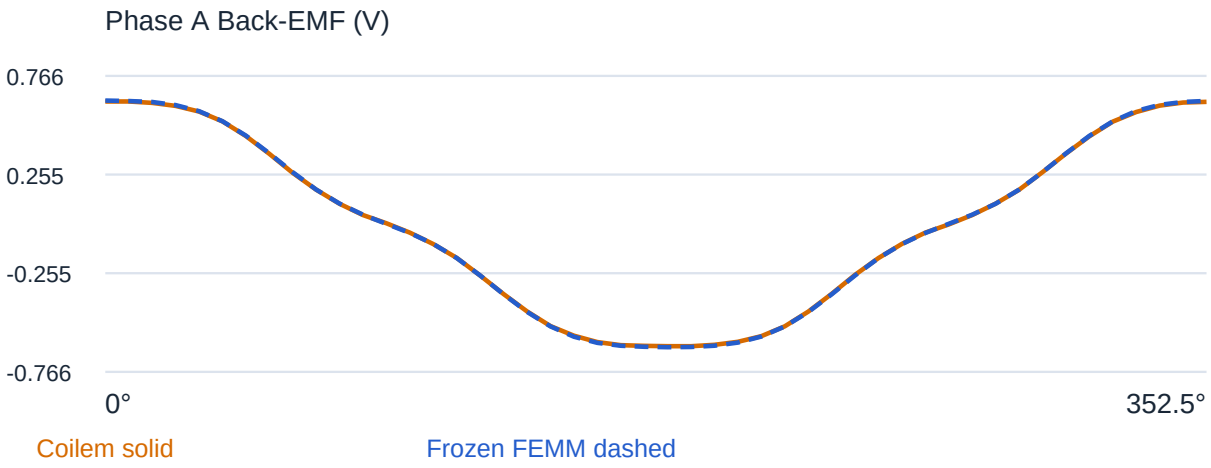
Residual solid



Coilem solid Frozen FEMM dashed



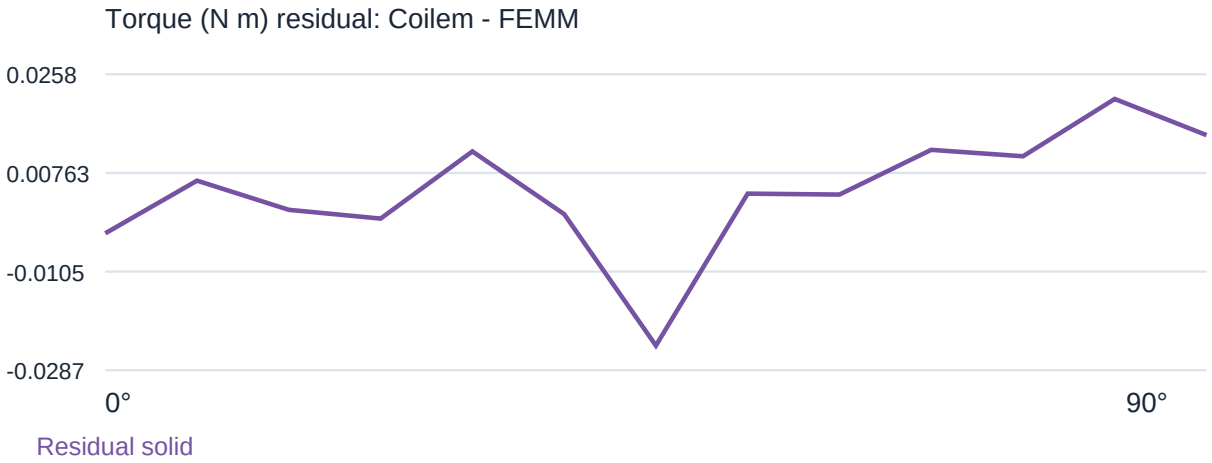
Residual solid



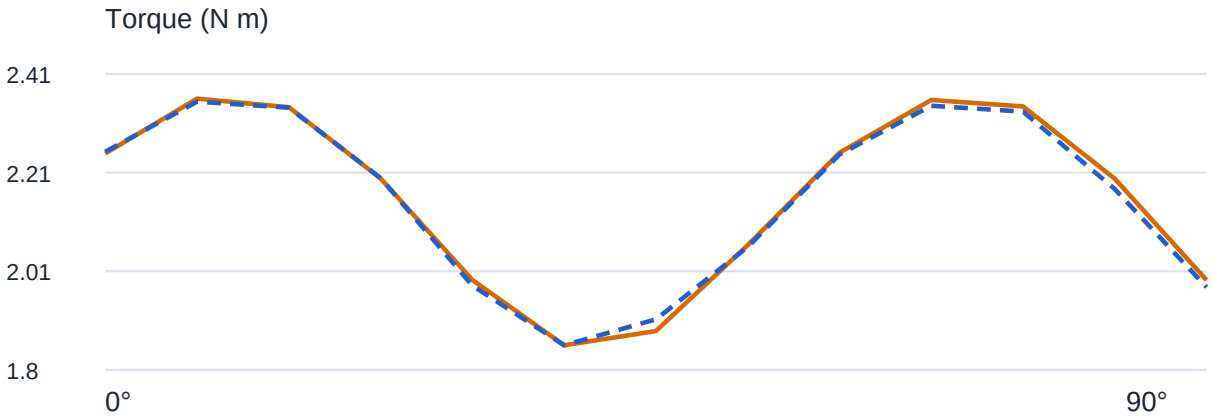
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_flat_8p12s_fsae_angled_pocket_contract

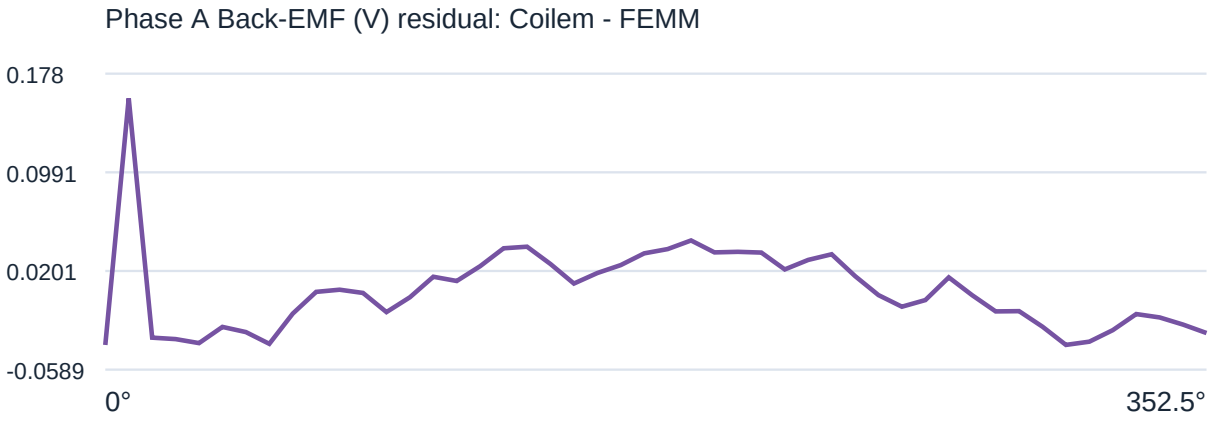
Frozen native fixture JSON



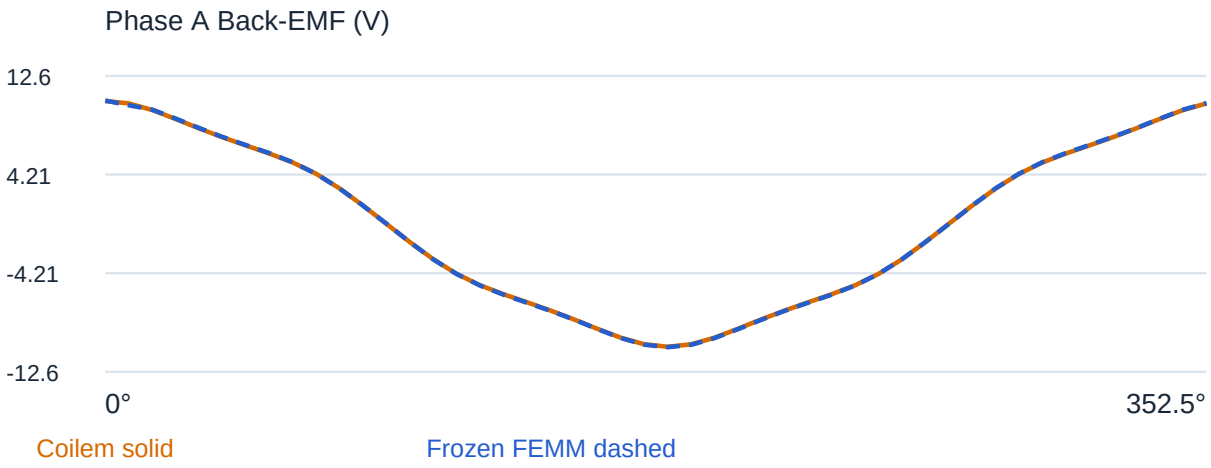
Residual solid



Coilem solid Frozen FEMM dashed



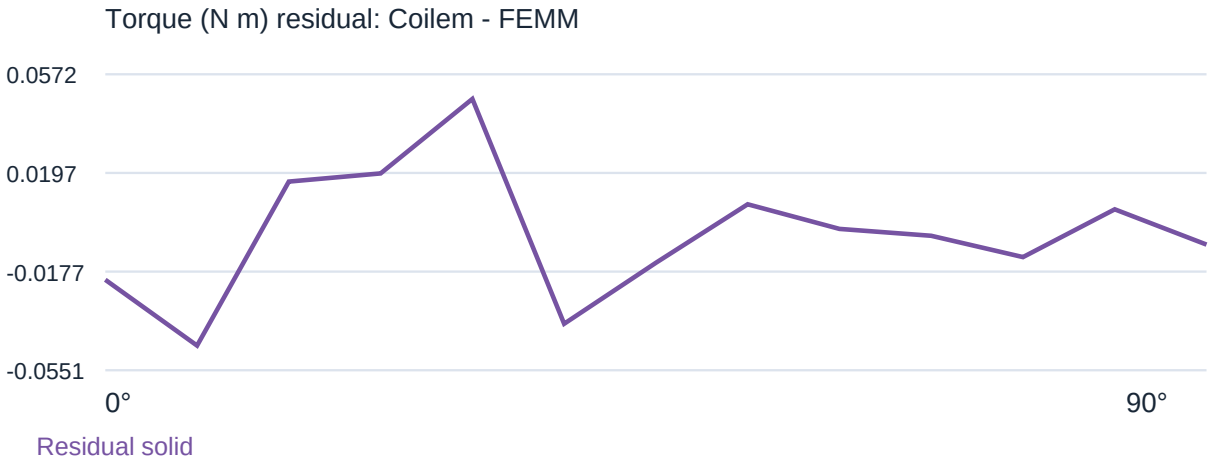
Residual solid



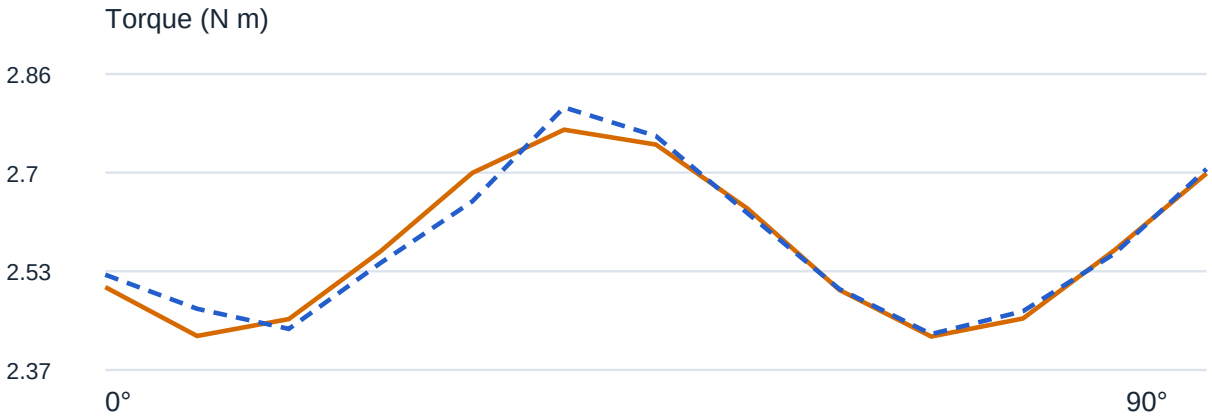
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_flat_8p12s_fsae_contract

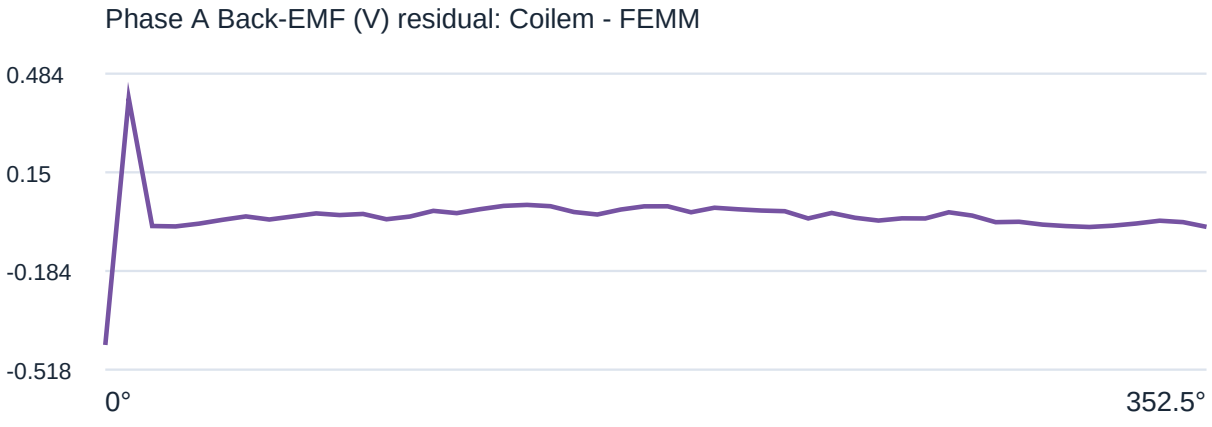
Frozen native fixture JSON



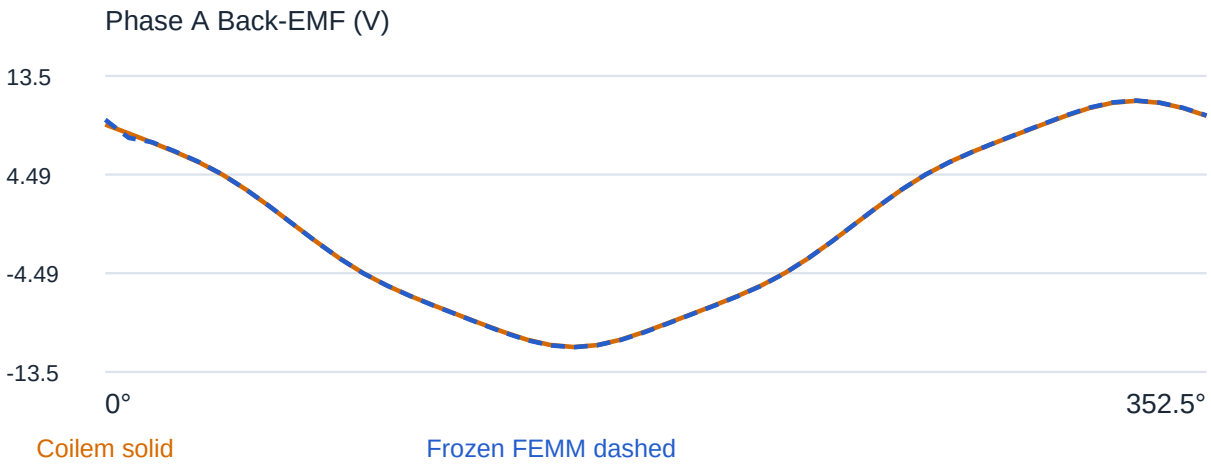
Residual solid



Coilem solid Frozen FEMM dashed



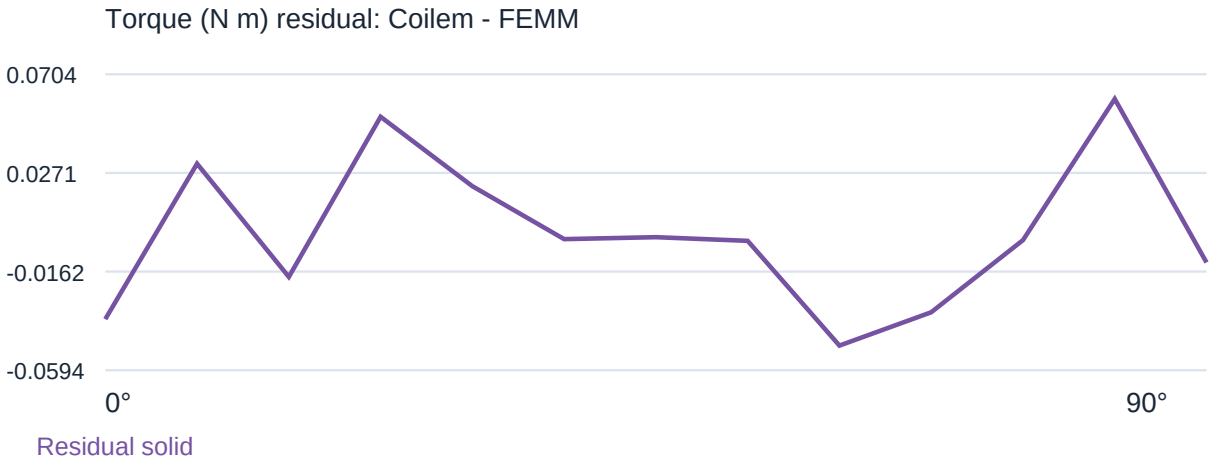
Residual solid



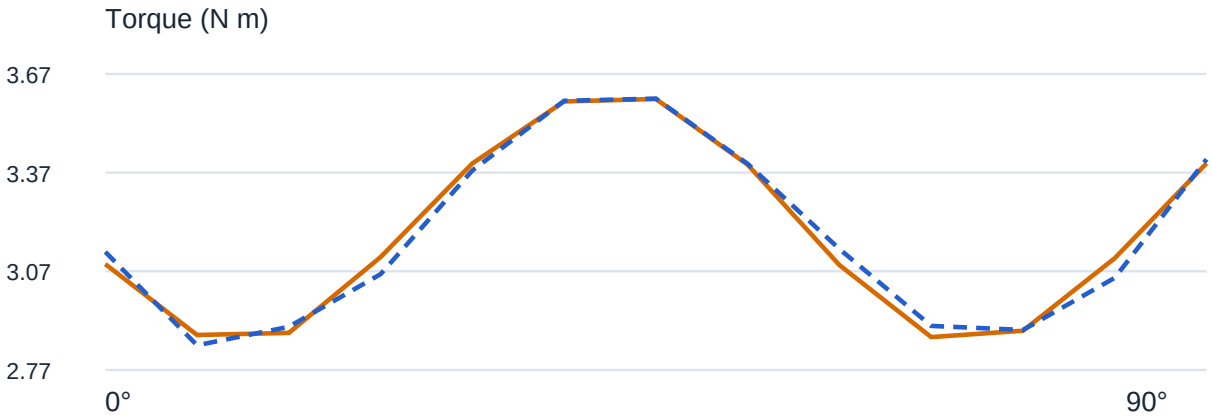
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_flat_8p12s_fsae_thin_bridge_contract

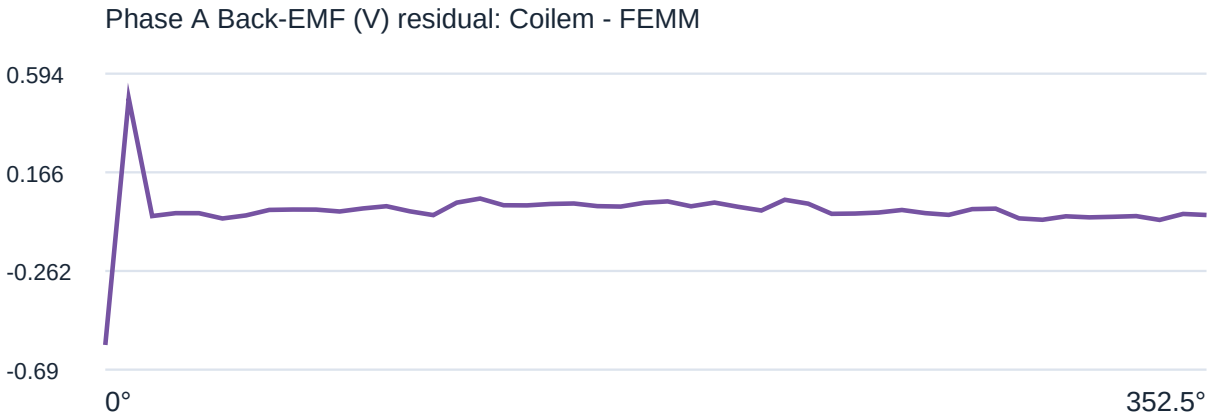
Frozen native fixture JSON



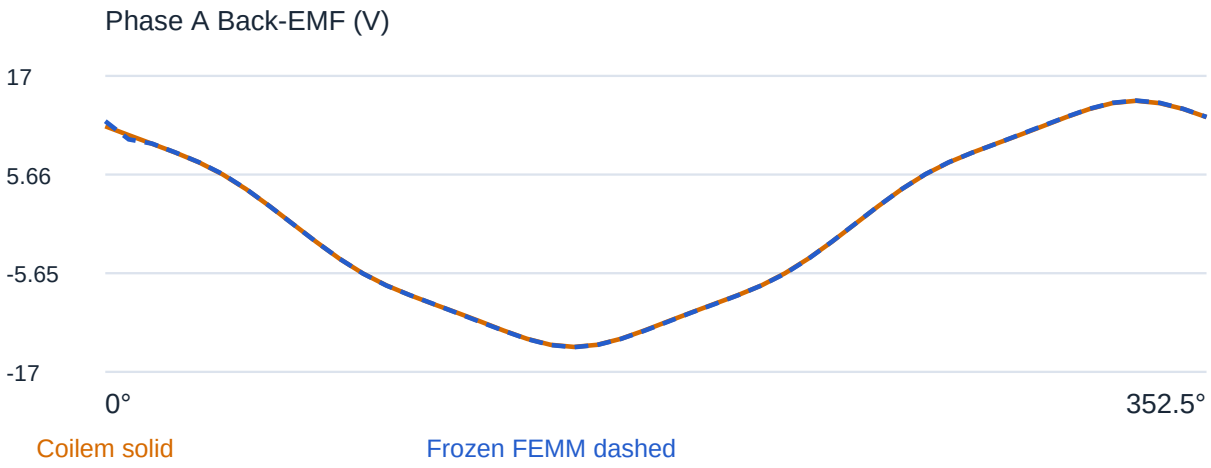
Residual solid



Coilem solid Frozen FEMM dashed



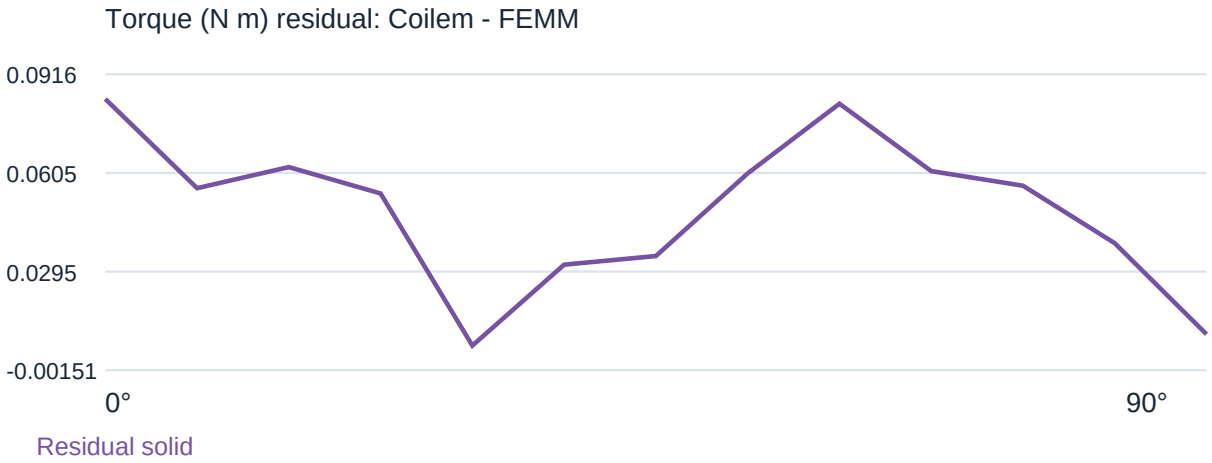
Residual solid



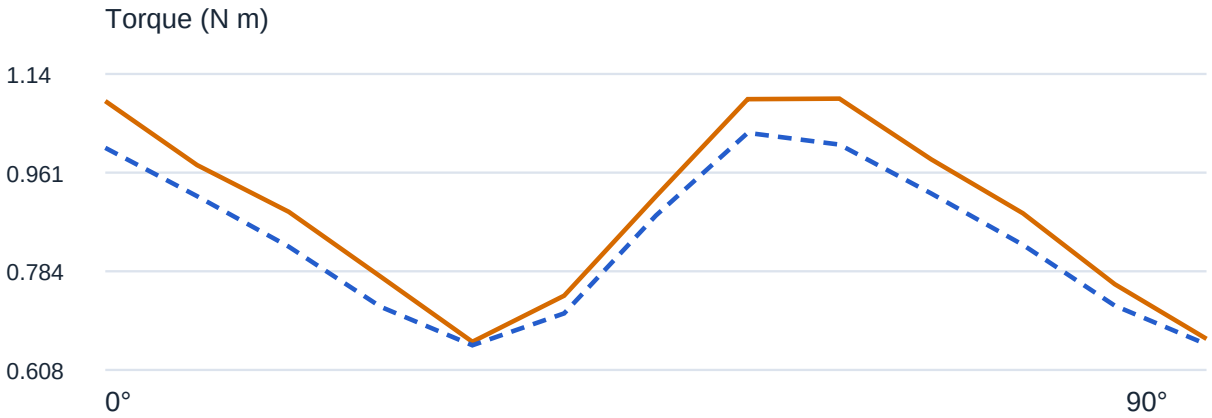
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_vshape_8p12s_baseline_contract

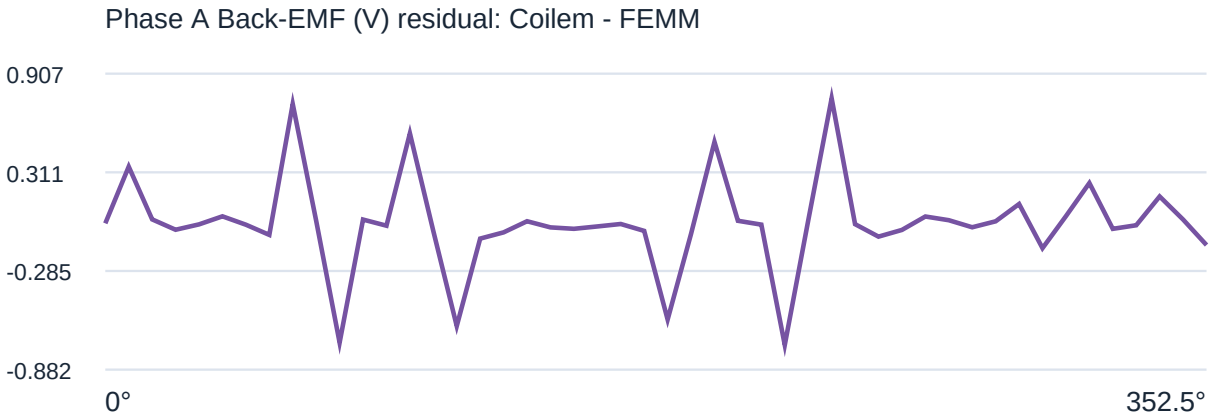
Frozen native fixture JSON



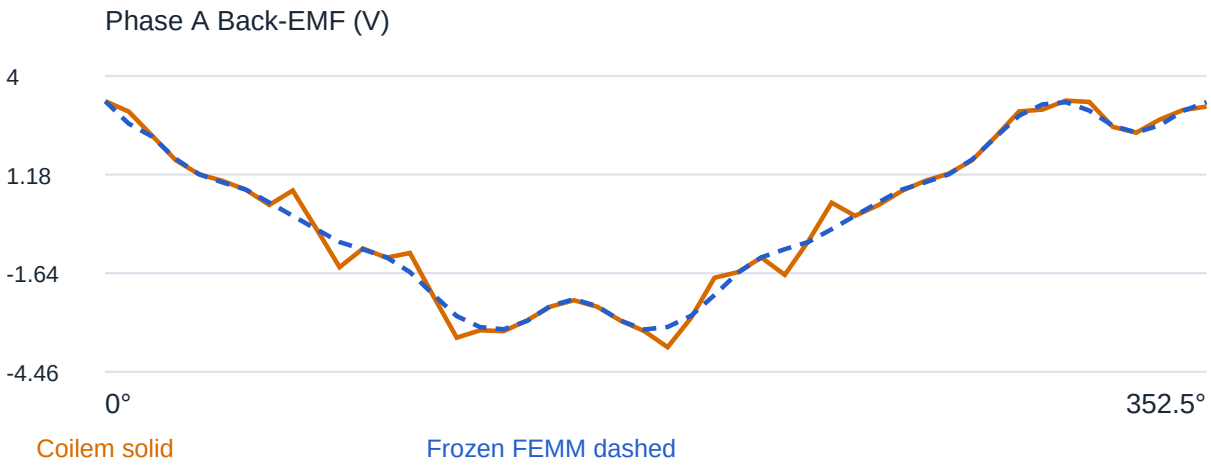
Residual solid



Coilem solid Frozen FEMM dashed



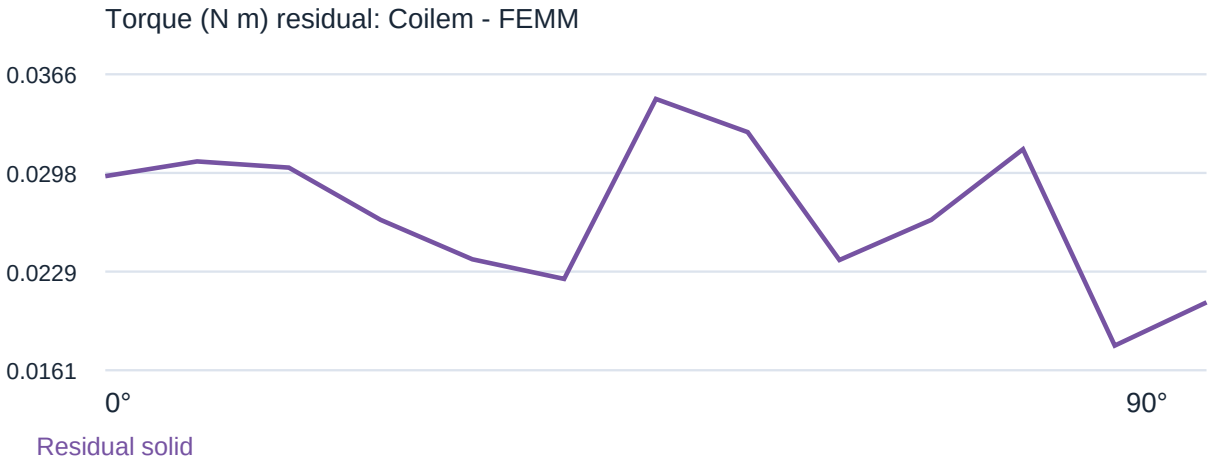
Residual solid

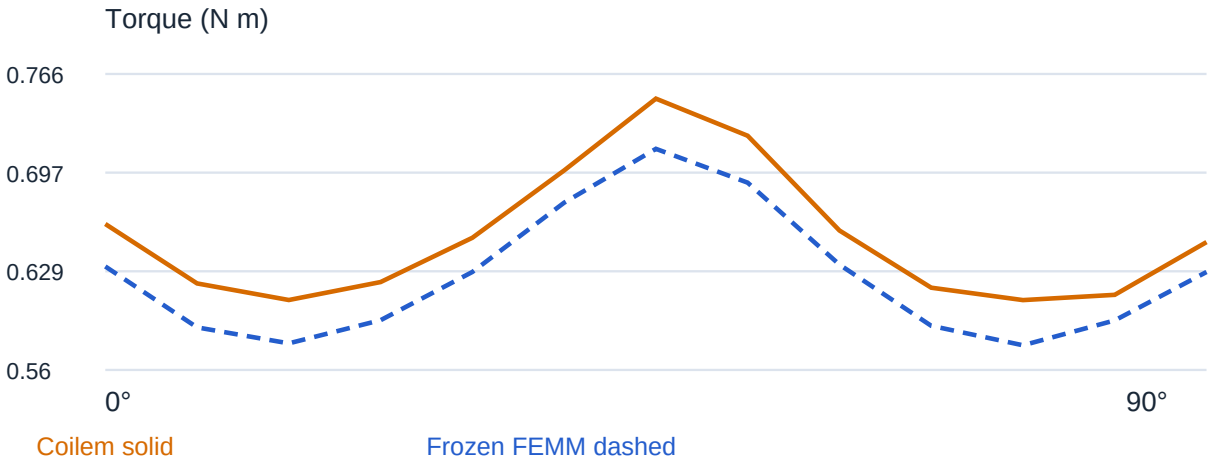


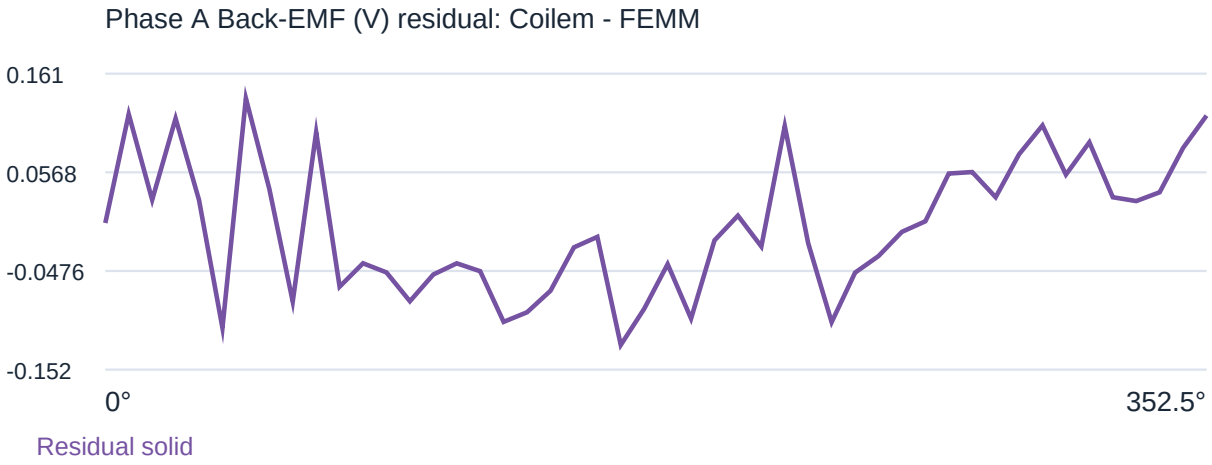
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	YELLOW
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

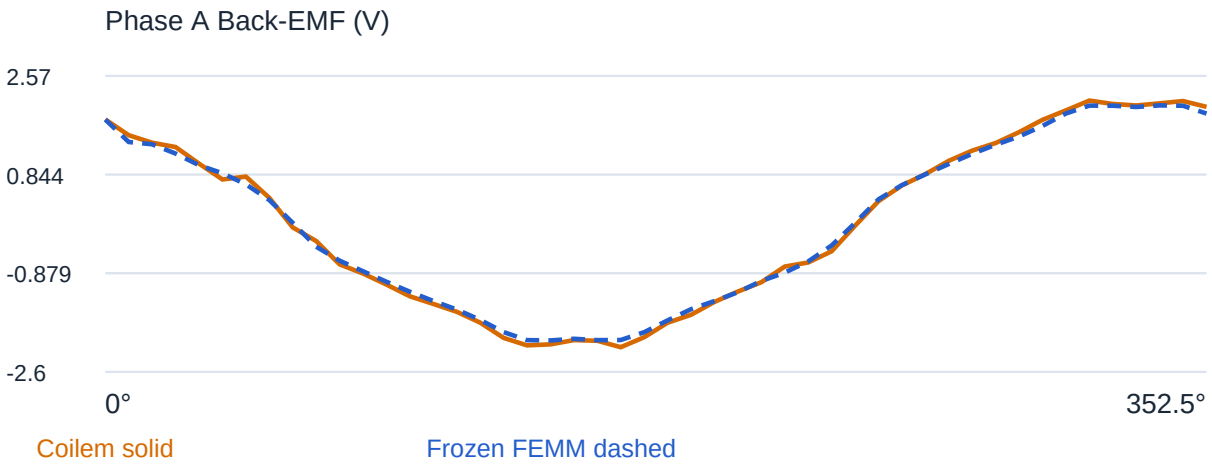
Phase B – ipm_vshape_8p12s_deep_apex_high_output_v1

Frozen native fixture JSON





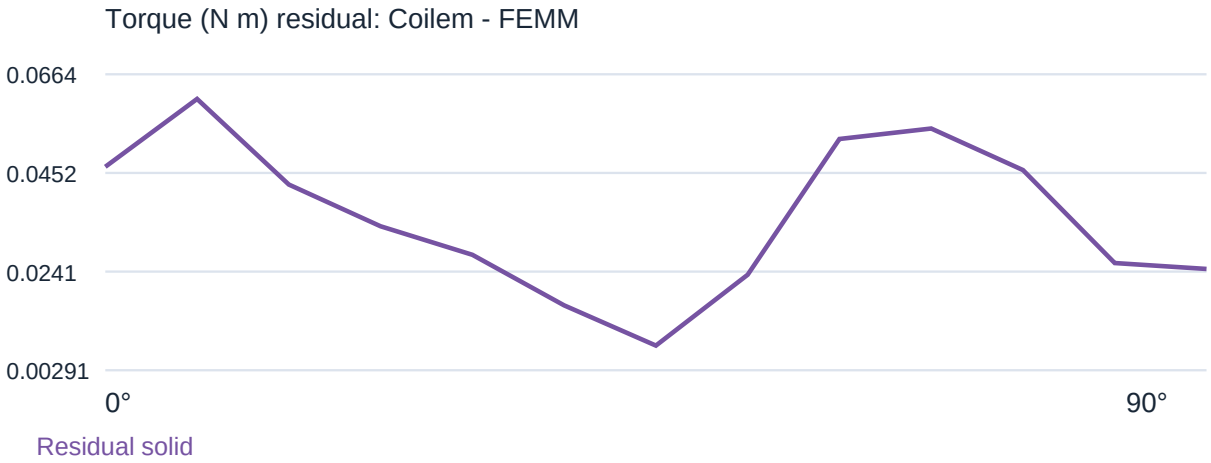




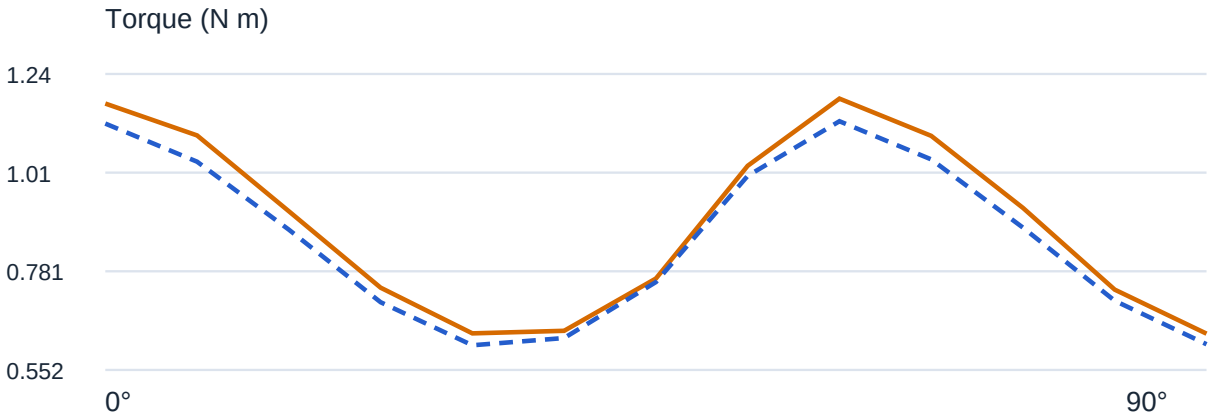
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_vshape_8p12s_narrow_angle_contract

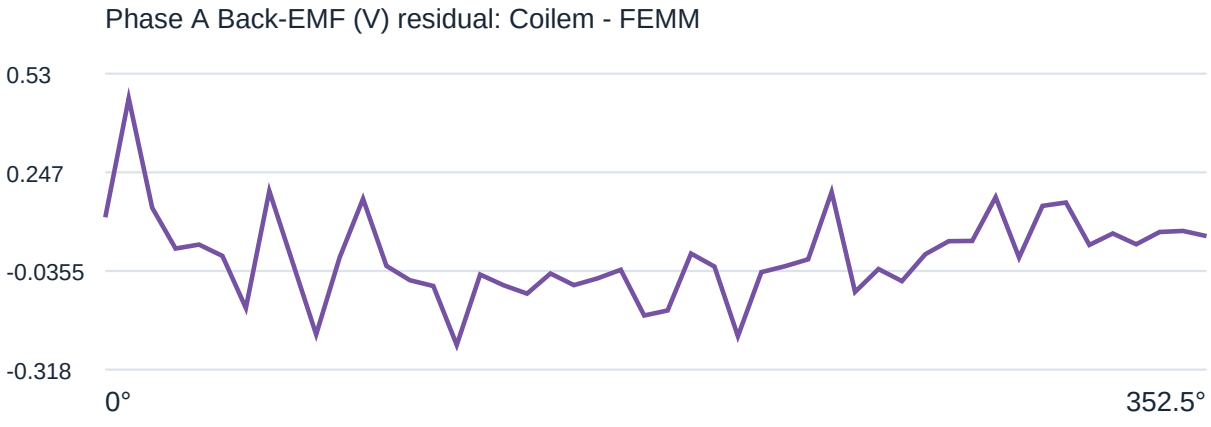
Frozen native fixture JSON



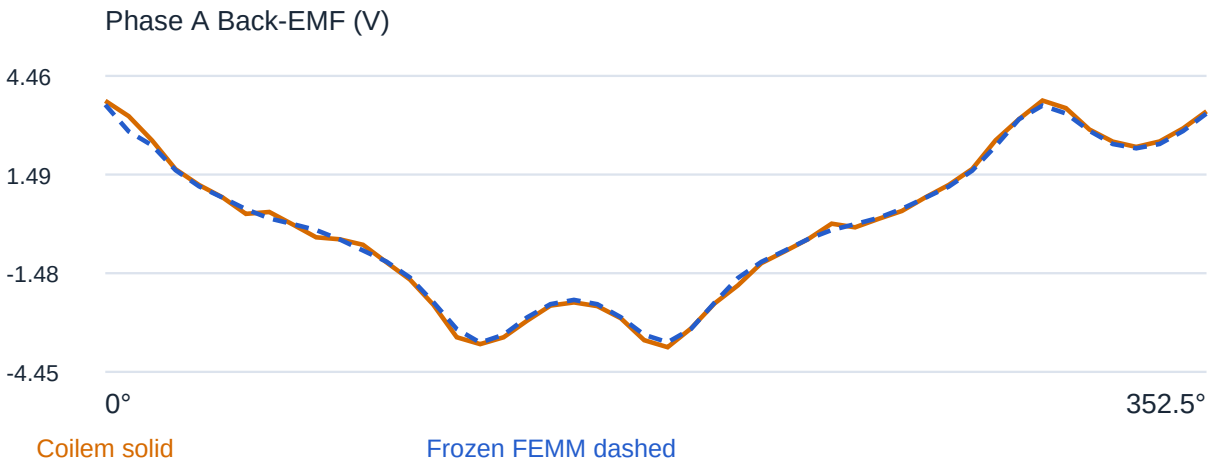
Residual solid



Coilem solid Frozen FEMM dashed



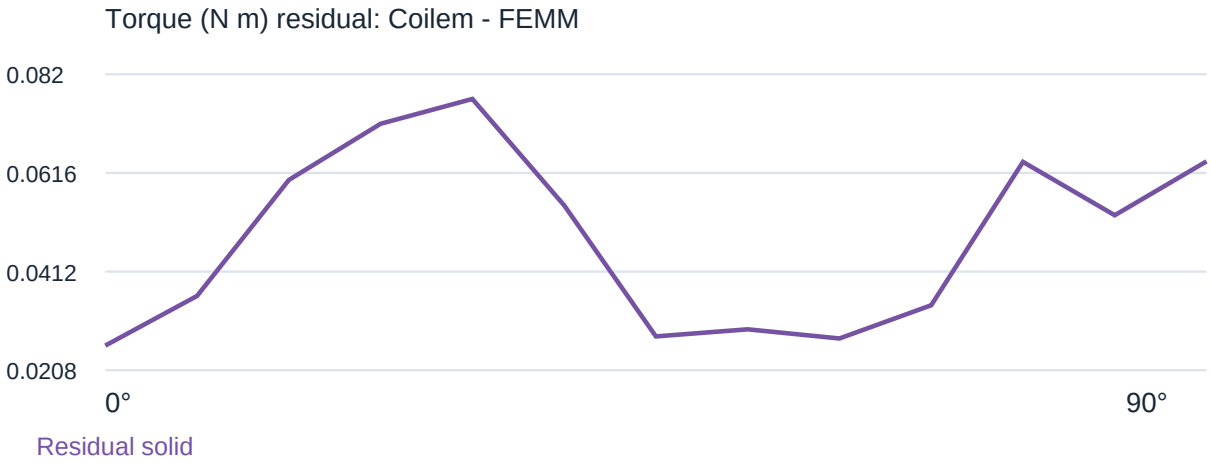
Residual solid



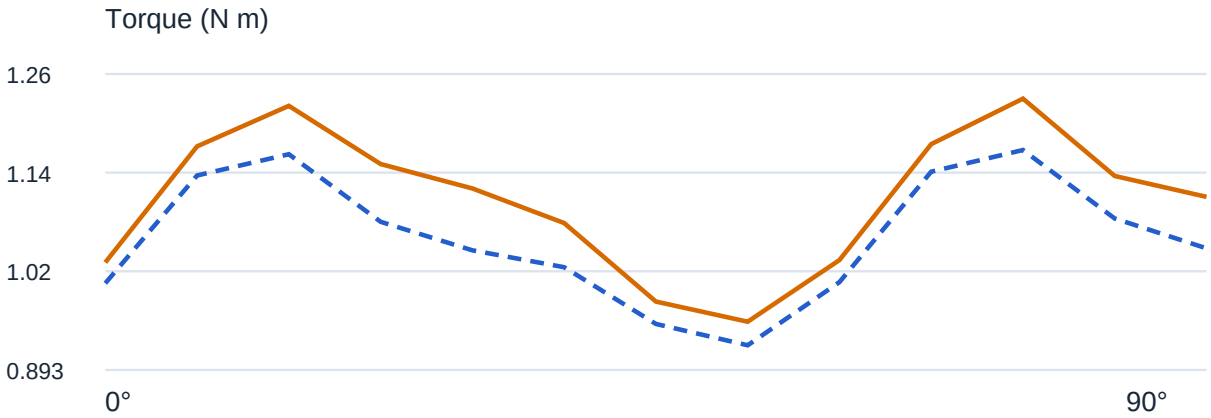
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_vshape_8p12s_pocket_offset_contract

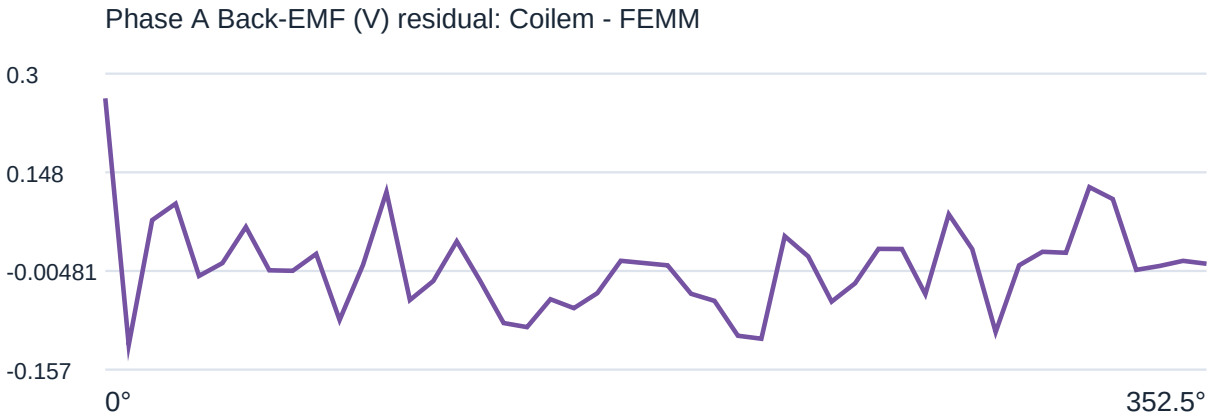
Frozen native fixture JSON



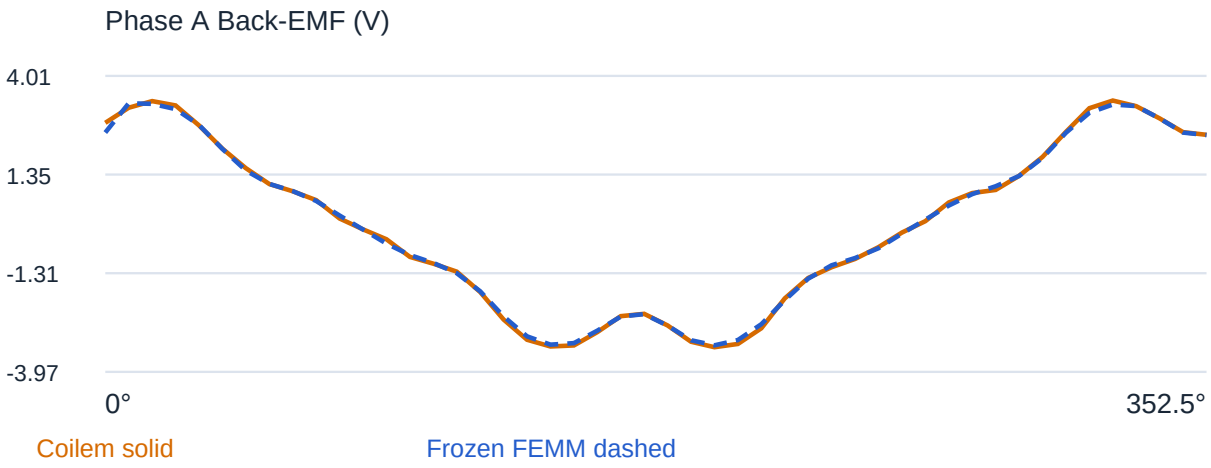
Residual solid



Coilem solid Frozen FEMM dashed



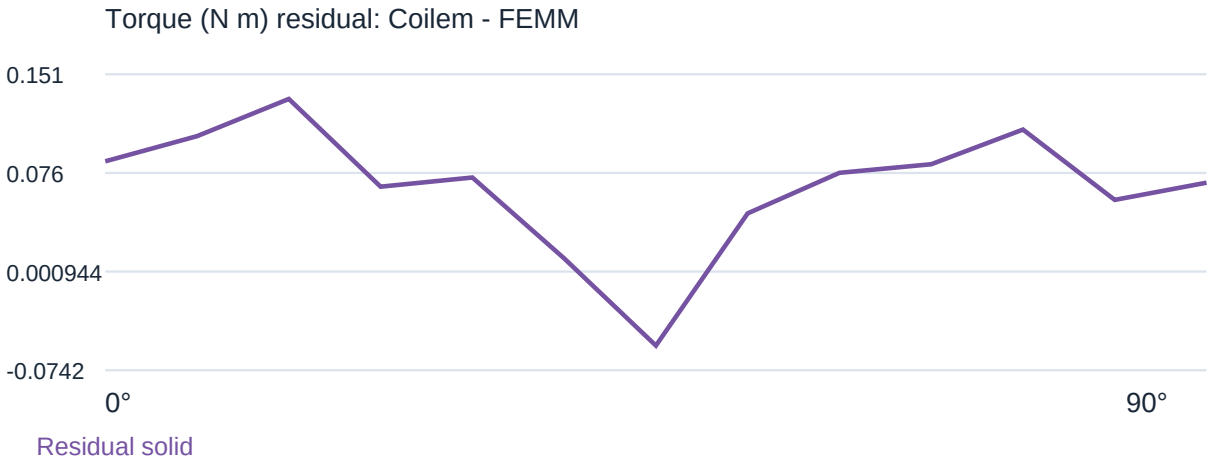
Residual solid

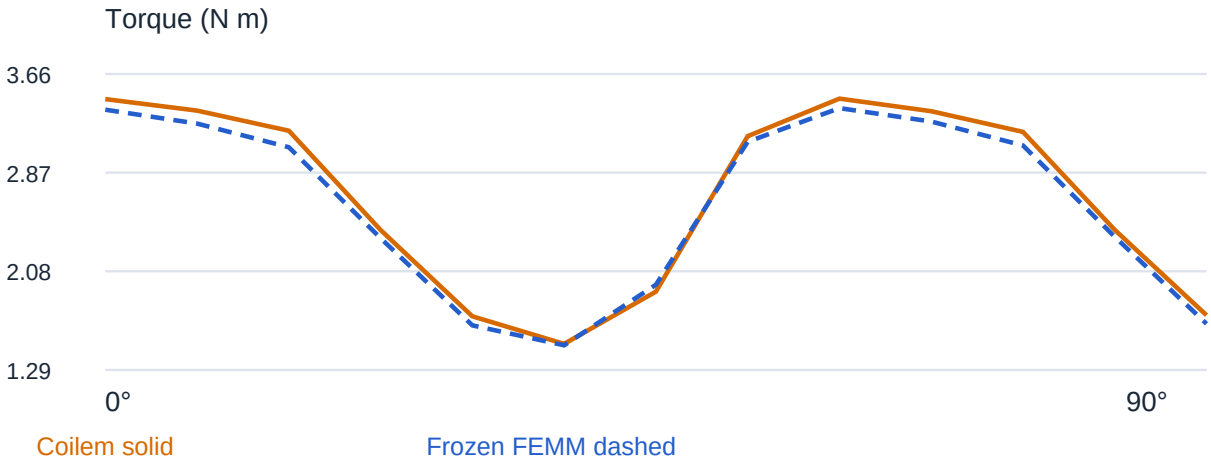


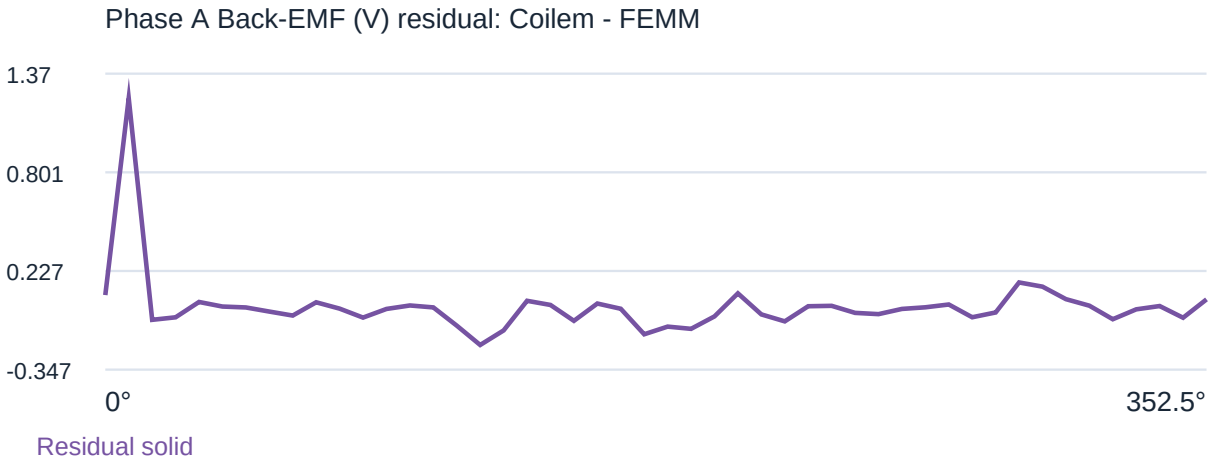
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

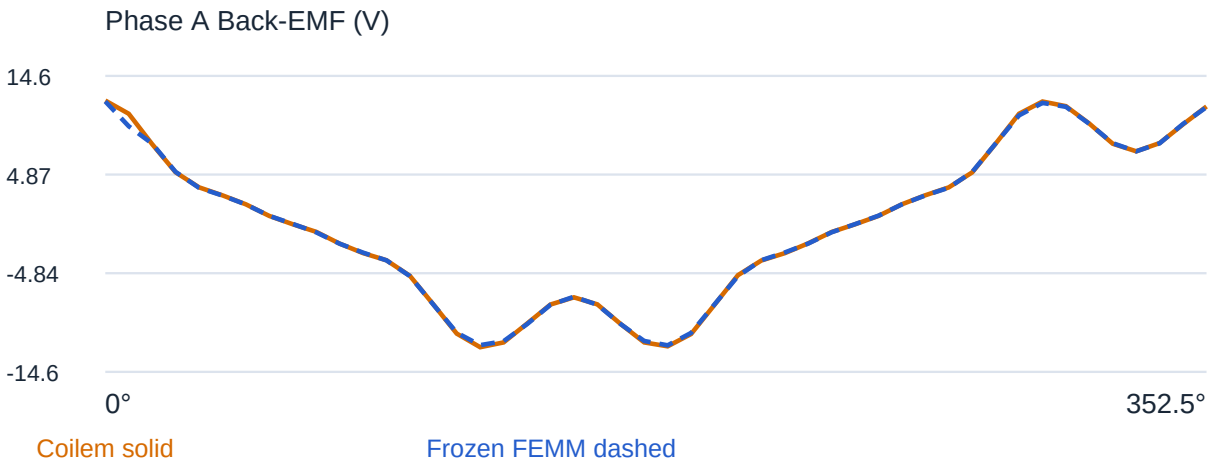
Phase B – ipm_vshape_8p12s_thin_bridge_contract

Frozen native fixture JSON





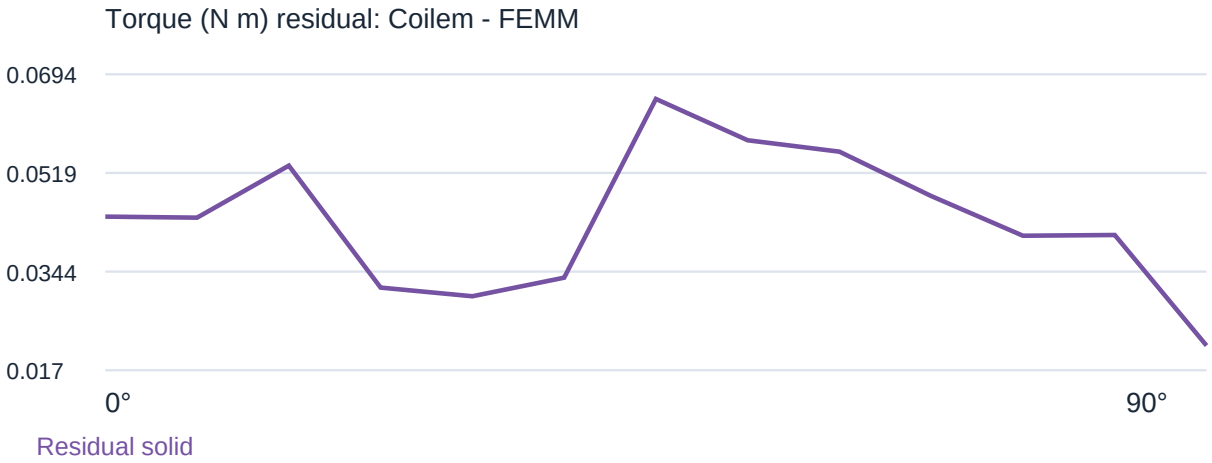




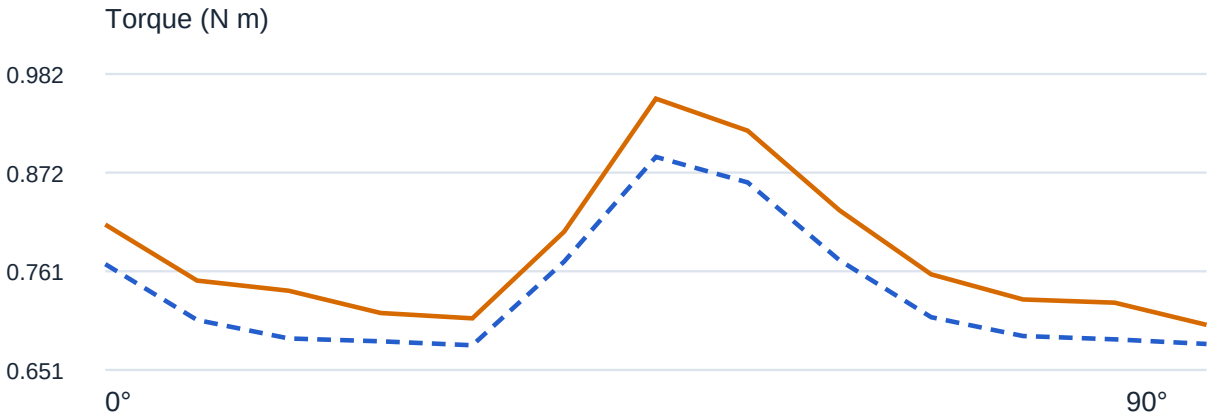
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase B – ipm_vshape_8p12s_wide_angle_high_output_v1

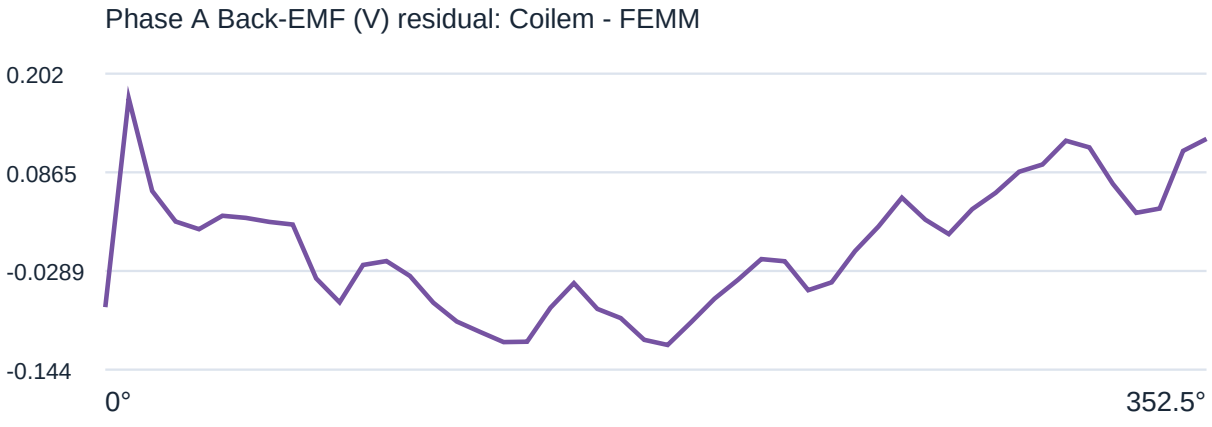
Frozen native fixture JSON



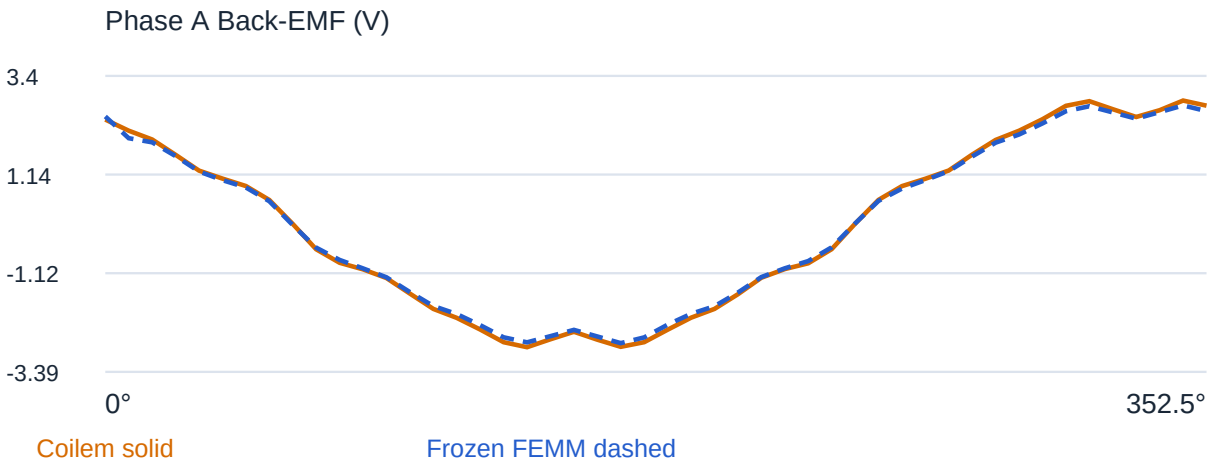
Residual solid



Coilem solid Frozen FEMM dashed



Residual solid



Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase C – ornl_prius_2004_ipm

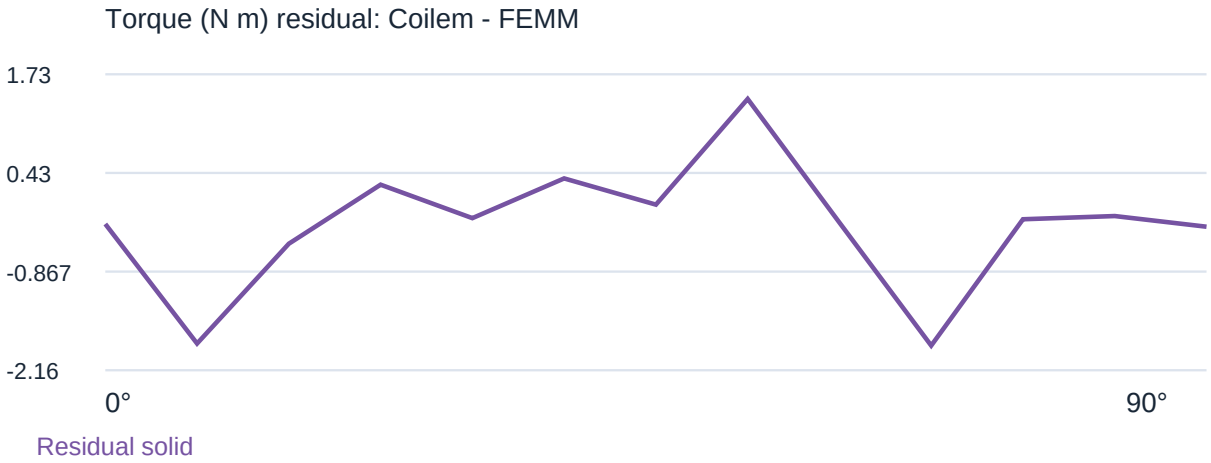
Frozen native fixture JSON

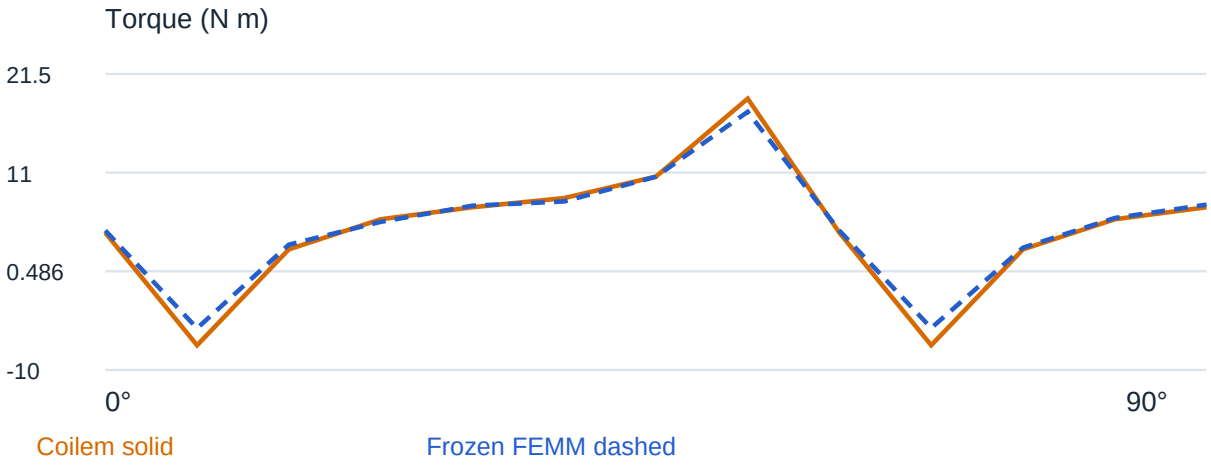
Planned coverage (not yet supported). The Prius fixture uses HoleM50 rotor geometry; the public API currently accepts flat-buried and V-shape IPM only. Qualify the Prius geometry, native mesh and field solves against matched FEMM evidence before enabling public support. Numerical status: UNSUPPORTED; no comparison was performed.

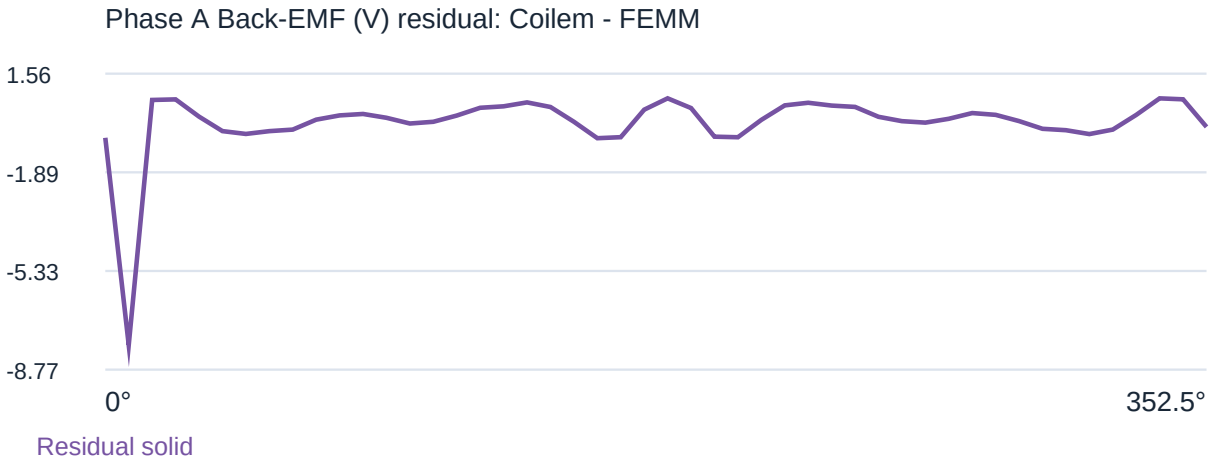
The public launch supports flat-buried and V-shape IPM only.

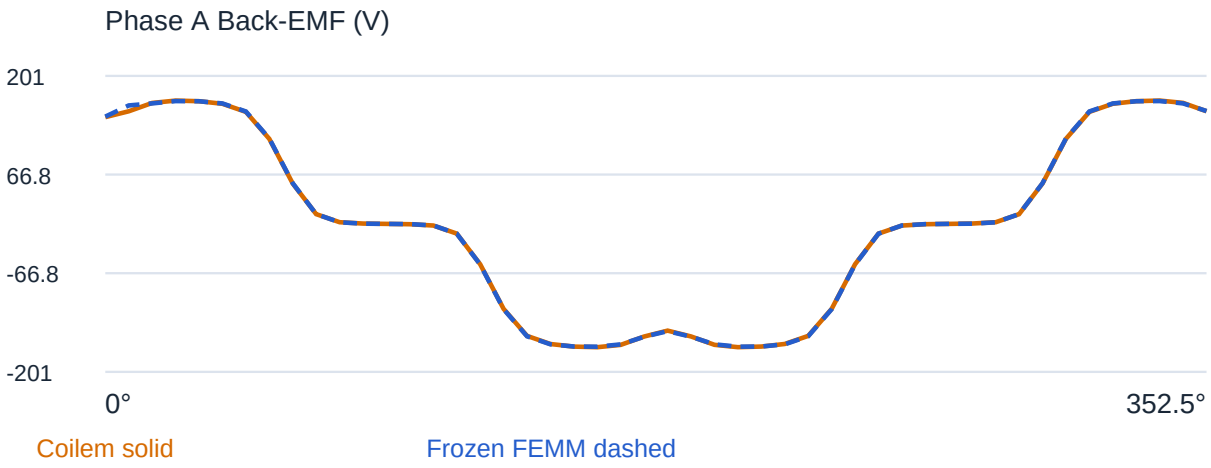
Phase C – spm_12p36s_distributed_femm_validation

Frozen native fixture JSON





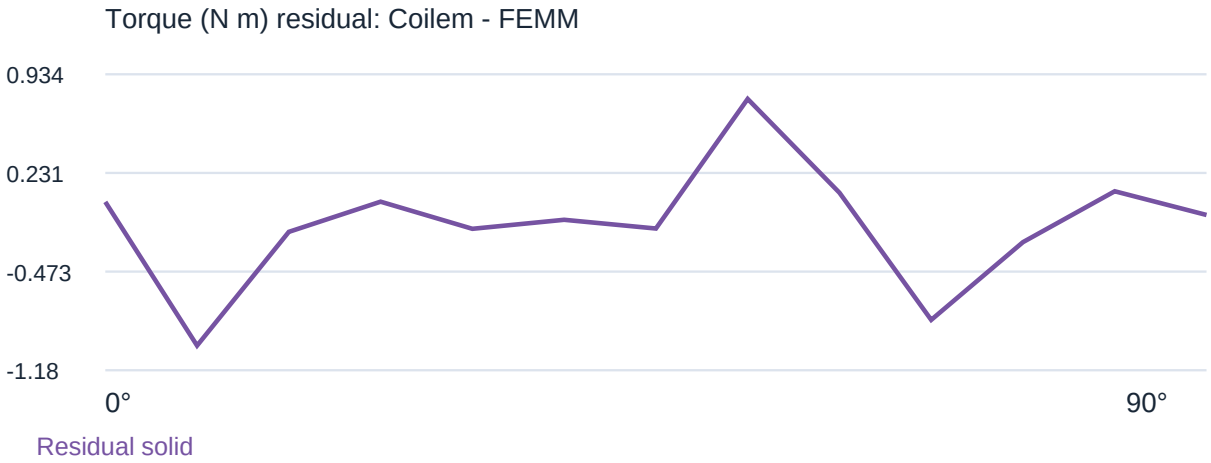


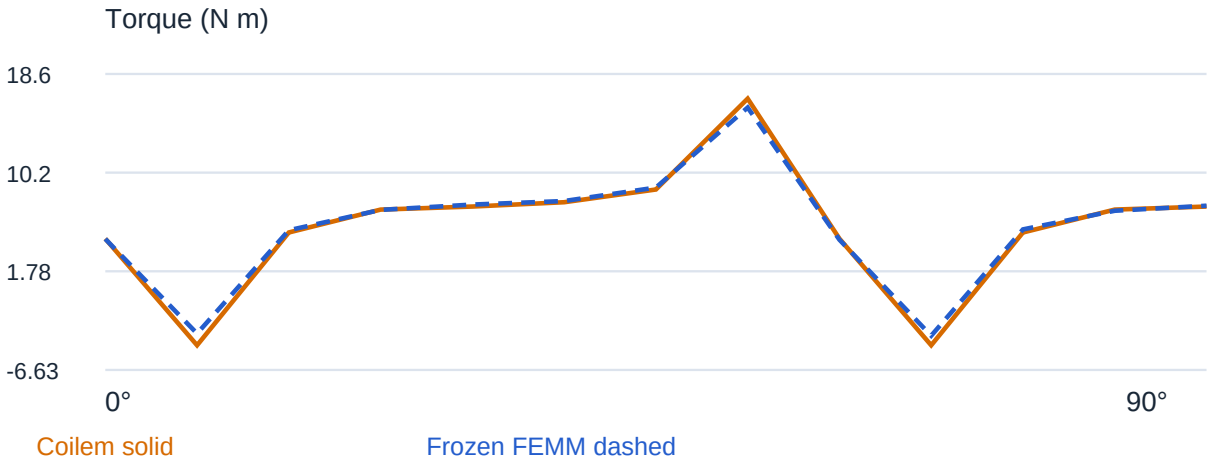


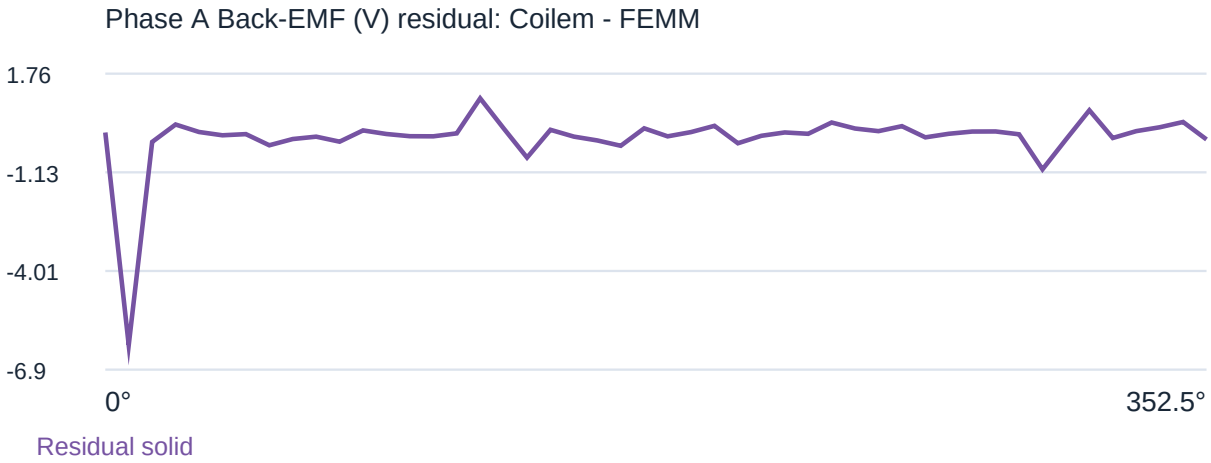
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

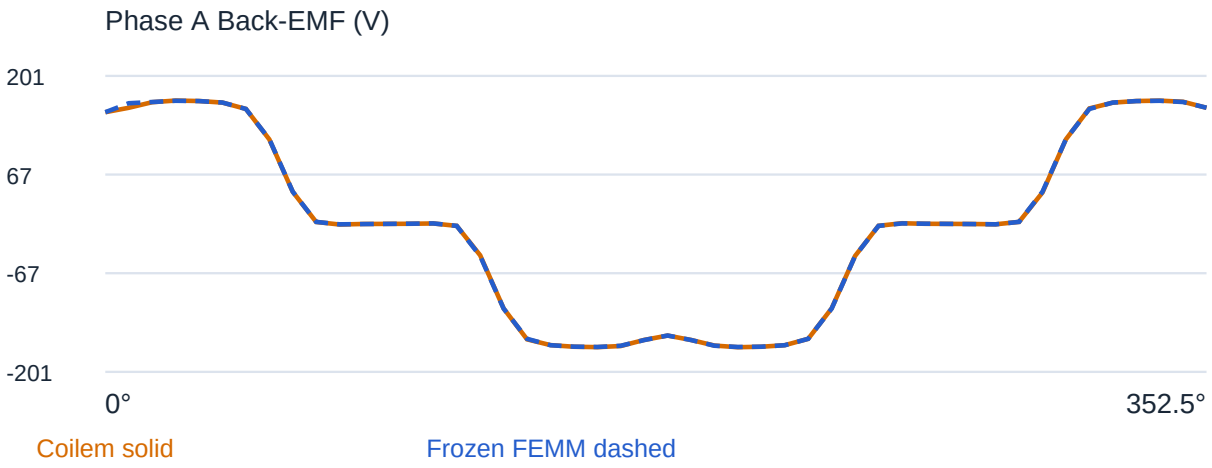
Phase C – spm_8p24s_distributed_femm_validation

Frozen native fixture JSON





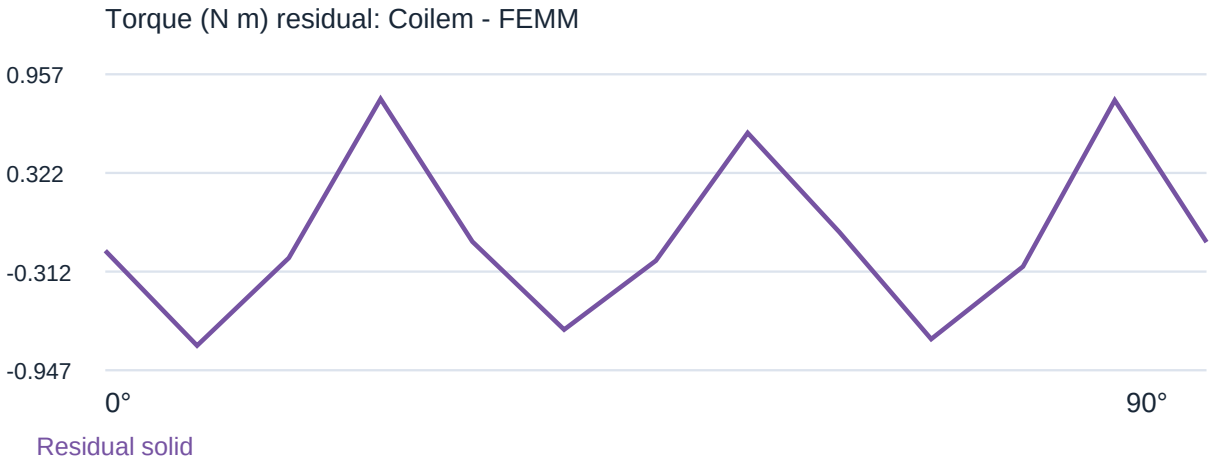




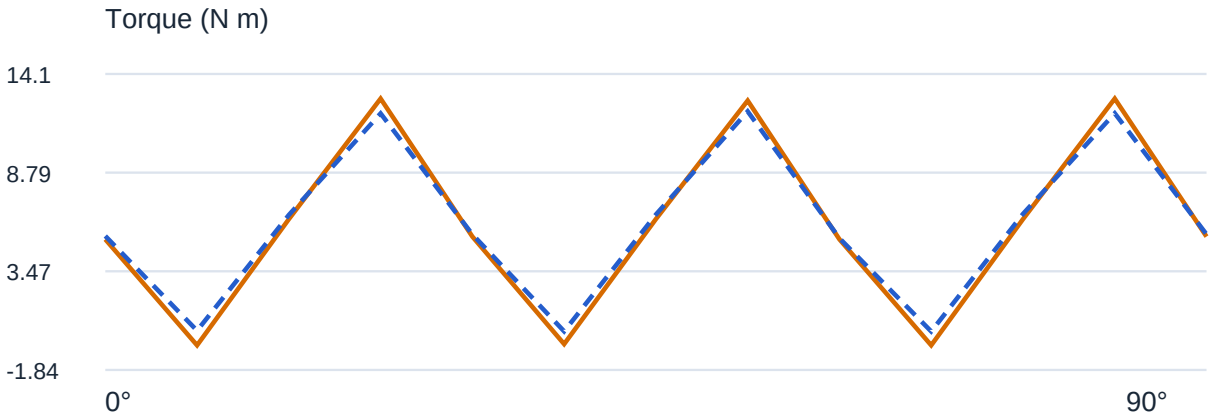
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase C – spm_8p48s_chorded_5_6

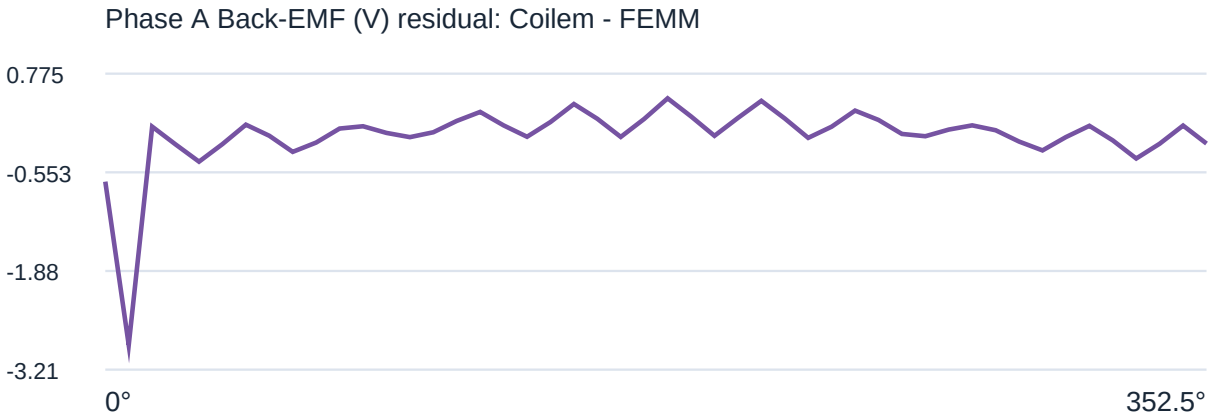
Frozen native fixture JSON



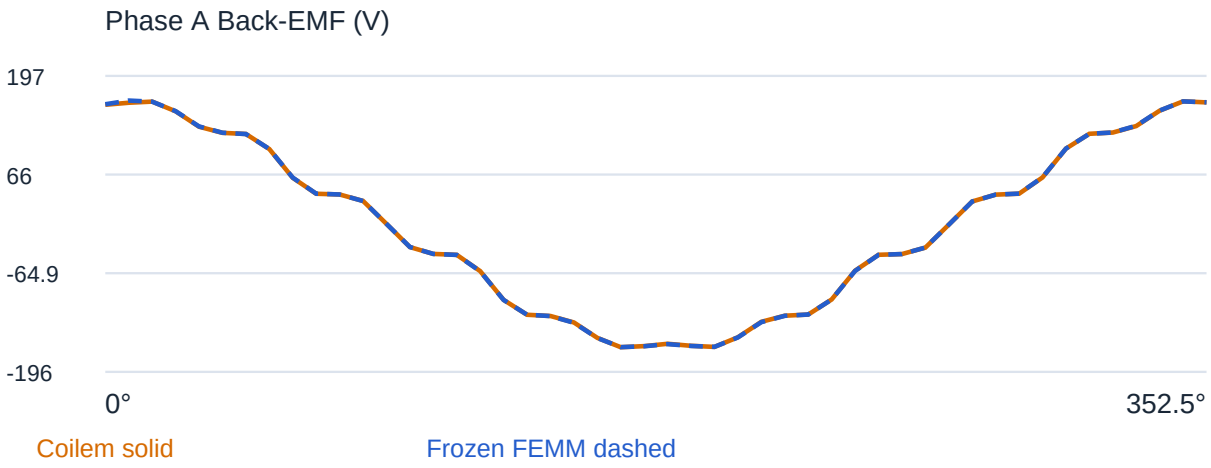
Residual solid



Coilem solid Frozen FEMM dashed



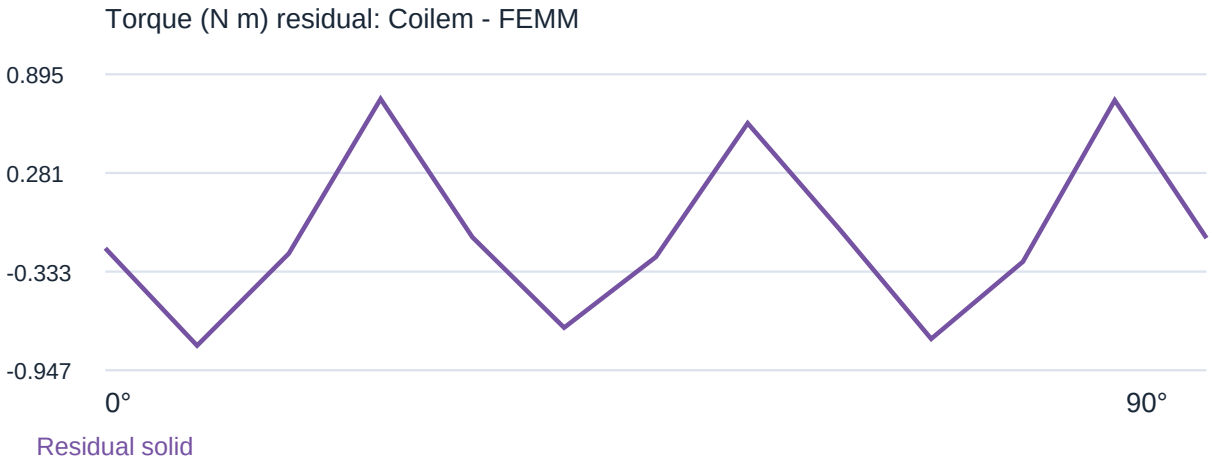
Residual solid



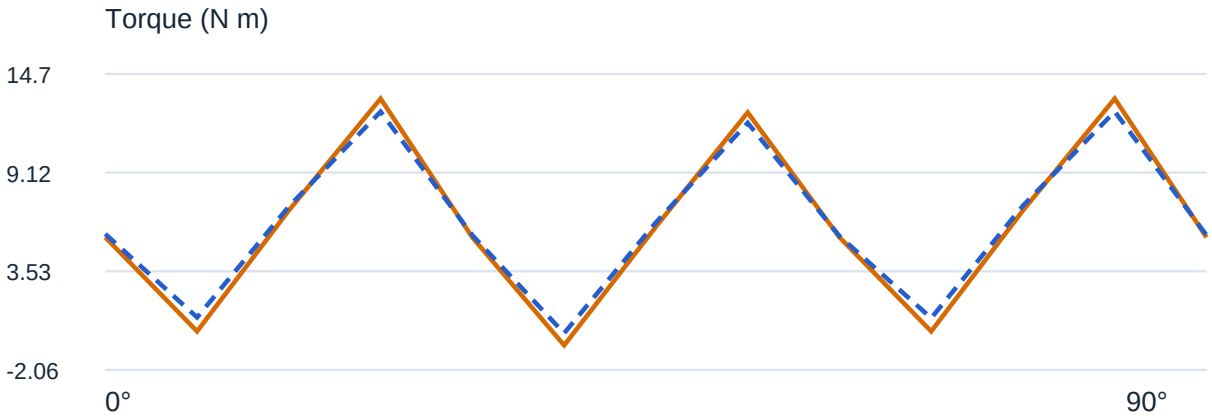
Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase C – spm_8p48s_distributed_femm_validation

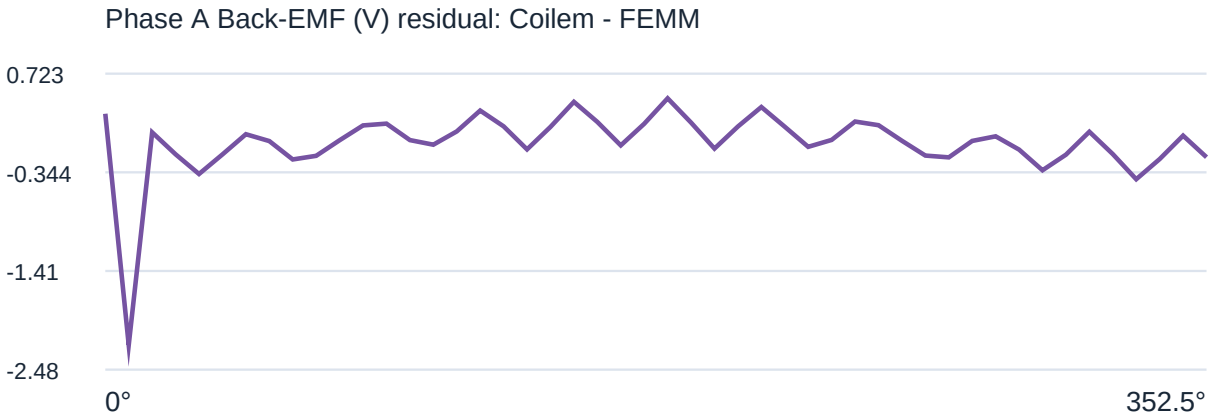
Frozen native fixture JSON



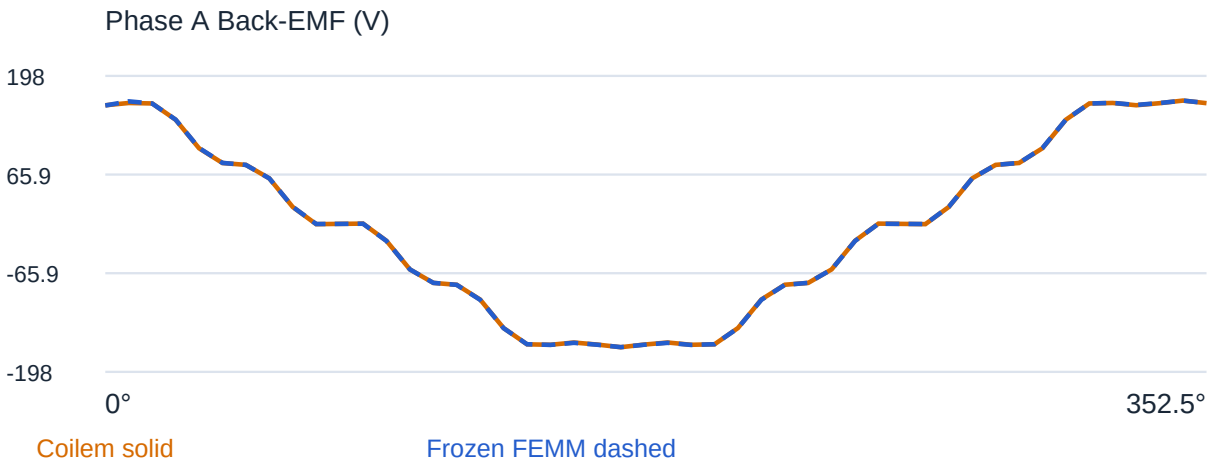
Residual solid



Coilem solid Frozen FEMM dashed



Residual solid



Gate	Outcome
mean_torque	PASS
bemf_fundamental	PASS
bemf_peak	PASS
torque_method	PASS
native_mesh	PASS
angle_grids	PASS

Phase D – BLDC results

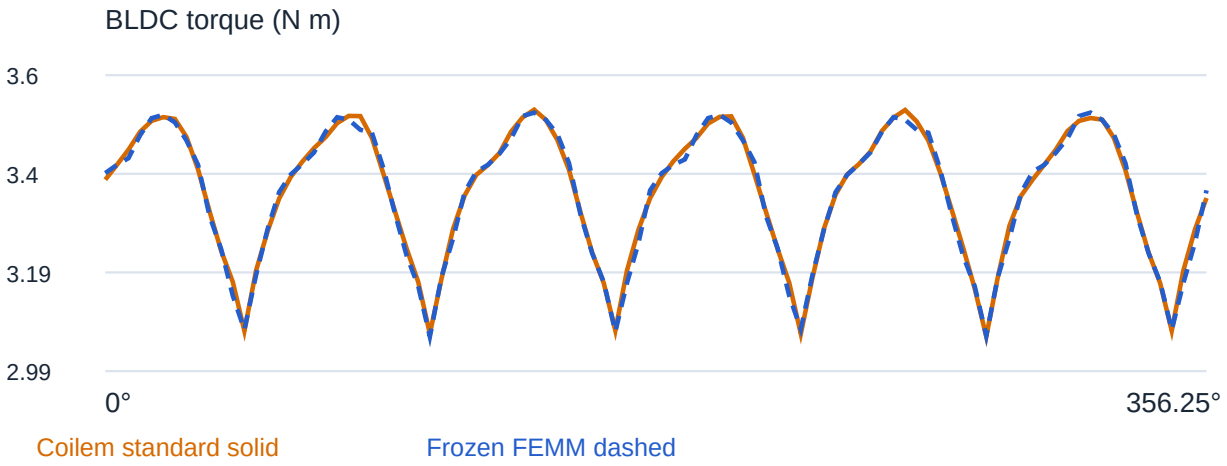
Fixture	Outcome
spm_14p12s_high_pole	PASS
spm_4p12s_small	PASS
spm_8p12s_reference	PASS

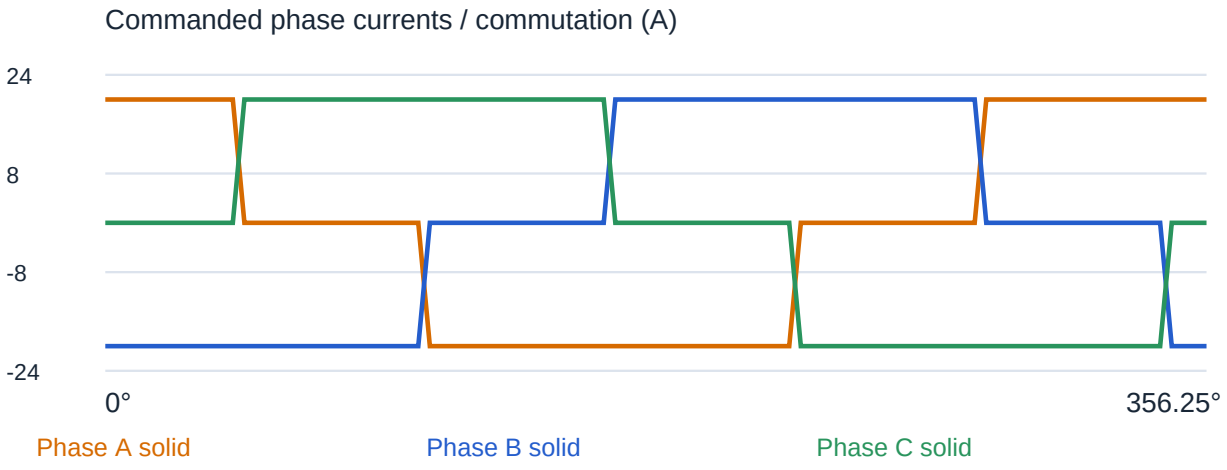
Phase D – spm_14p12s_high_pole

Frozen native-standard fixture JSON

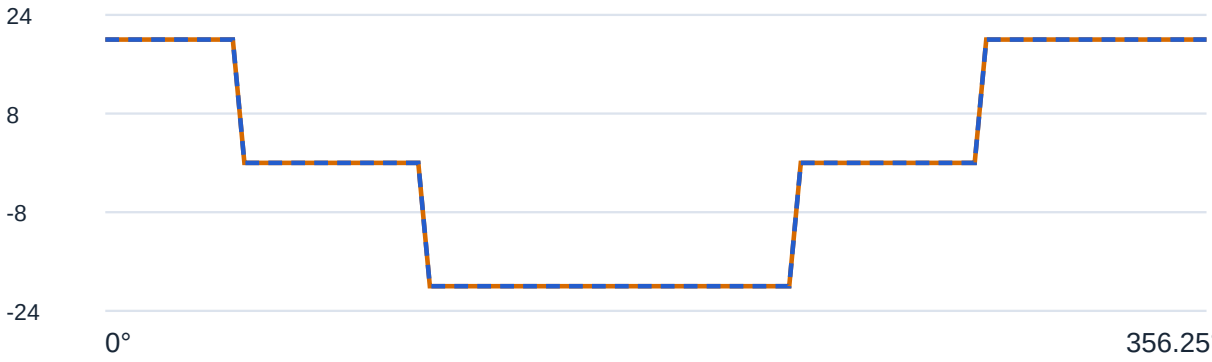
Frozen native-fine fixture JSON

Frozen analytical fixture JSON



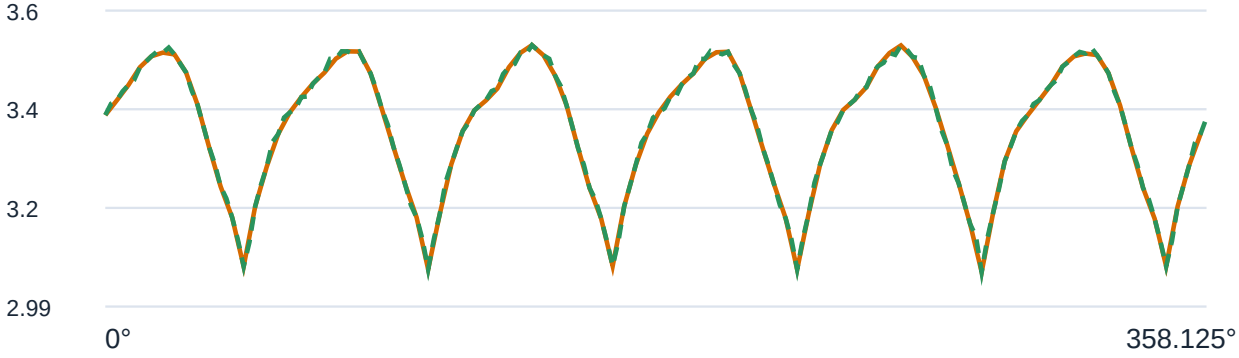


Phase A current: native / frozen reference (A)



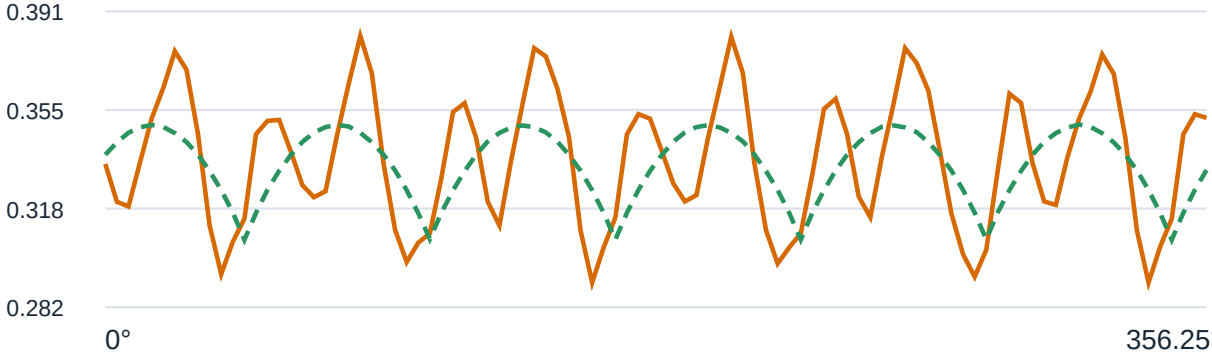
Coilem solid Frozen FEMM dashed

Angular convergence: normal spatial mesh (N m)



Standard solid Fine angular dashed

Low-current power / speed and field torque (N m); gate uses cycle means



Field torque solid Power / speed dashed

Gate	Outcome
mean_torque_delta_pct	PASS
torque_waveform_nrmse_pct	PASS
commutation_transition_delta_samples	PASS
power_torque_delta_pct	PASS
native_standard_to_fine_mean_torque_delta_pct	PASS
native_standard_to_fine_ripple_delta_percentage_points	PASS

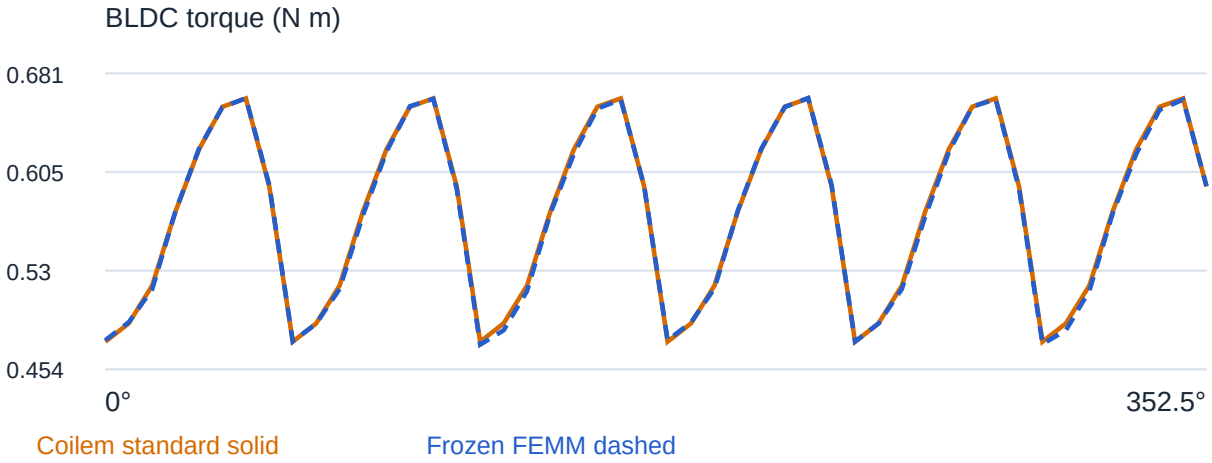
Gate	Outcome
native_standard_current_properties	PASS
native_fine_current_properties	PASS
femm_standard_current_properties	PASS
analytical_low_current_current_properties	PASS
native-standard_golden_current_sequence	PASS
native-fine_golden_current_sequence	PASS
analytical_golden_current_sequence	PASS
femm-standard_golden_current_sequence	PASS

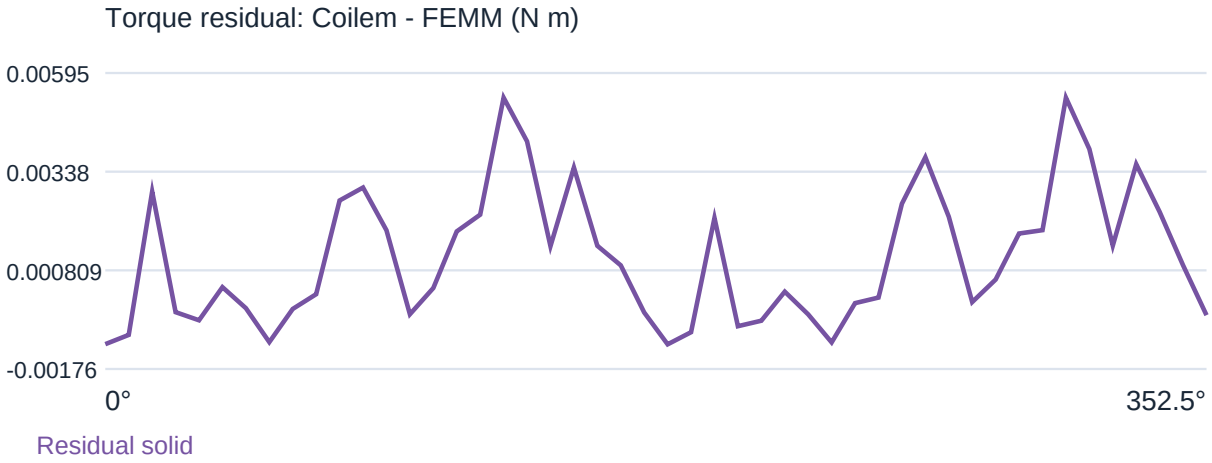
Phase D – spm_4p12s_small

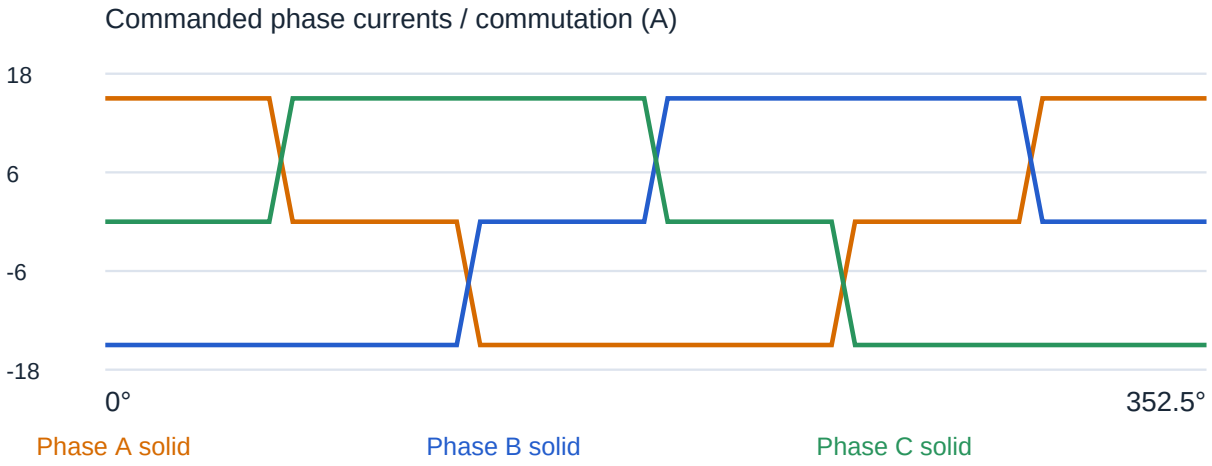
Frozen native-standard fixture JSON

Frozen native-fine fixture JSON

Frozen analytical fixture JSON





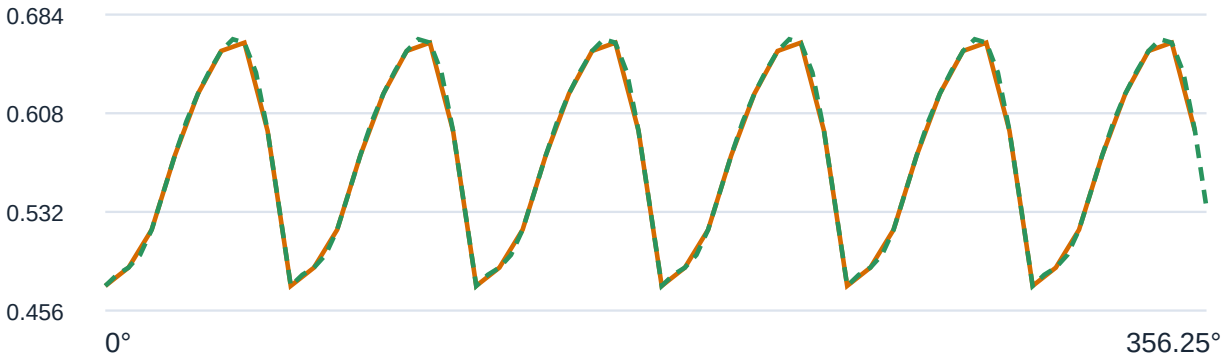


Phase A current: native / frozen reference (A)



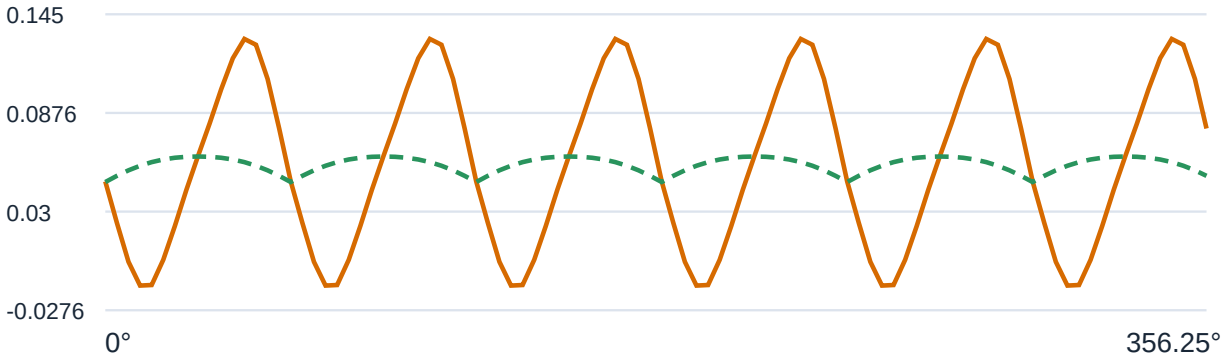
Coilem solid Frozen FEMM dashed

Angular convergence: normal spatial mesh (N m)



Standard solid Fine angular dashed

Low-current power / speed and field torque (N m); gate uses cycle means



Field torque solid Power / speed dashed

Gate	Outcome
mean_torque_delta_pct	PASS
torque_waveform_nrmse_pct	PASS
commutation_transition_delta_samples	PASS
power_torque_delta_pct	PASS
native_standard_to_fine_mean_torque_delta_pct	PASS
native_standard_to_fine_ripple_delta_percentage_points	PASS

Gate	Outcome
native_standard_current_properties	PASS
native_fine_current_properties	PASS
femm_standard_current_properties	PASS
analytical_low_current_current_properties	PASS
native-standard_golden_current_sequence	PASS
native-fine_golden_current_sequence	PASS
analytical_golden_current_sequence	PASS
femm-standard_golden_current_sequence	PASS

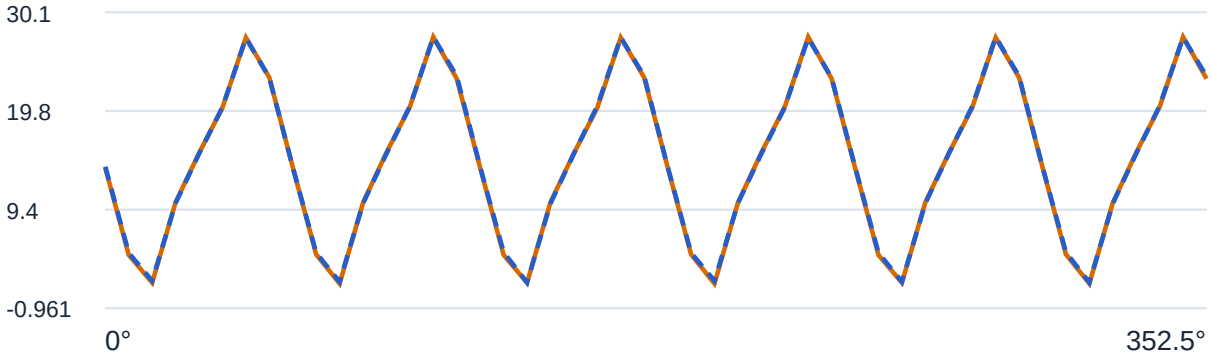
Phase D – spm_8p12s_reference

Frozen native-standard fixture JSON

Frozen native-fine fixture JSON

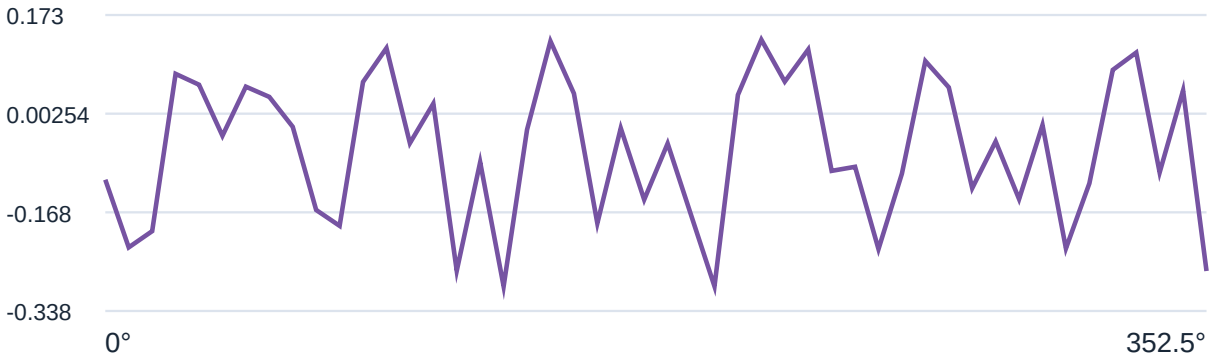
Frozen analytical fixture JSON

BLDC torque (N m)



Coilem standard solid Frozen FEMM dashed

Torque residual: Coilem - FEMM (N m)



Residual solid

Commanded phase currents / commutation (A)



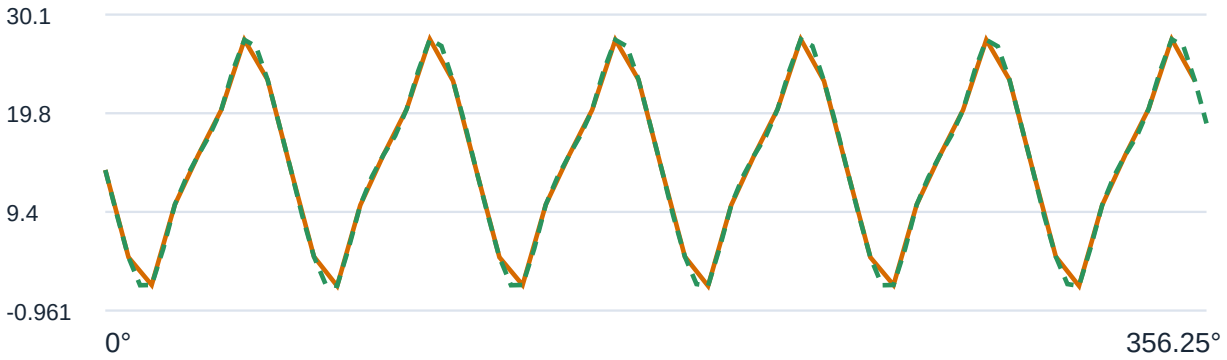
Phase A solid Phase B solid Phase C solid

Phase A current: native / frozen reference (A)



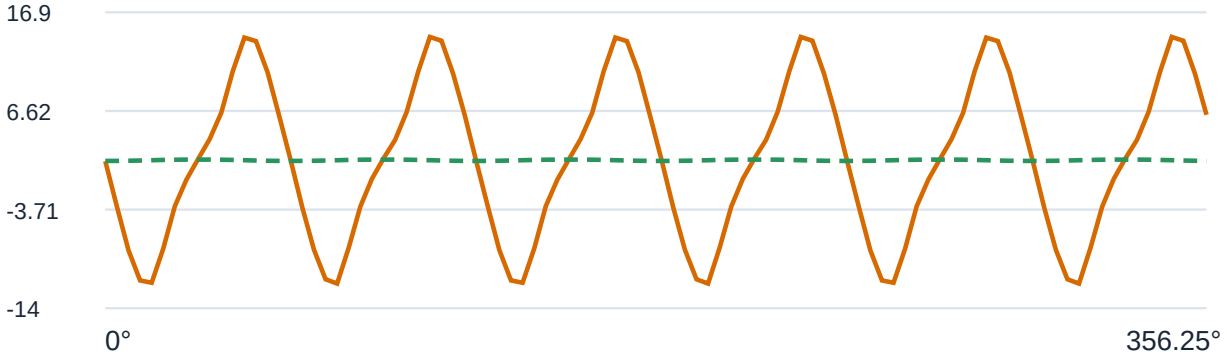
Coilem solid Frozen FEMM dashed

Angular convergence: normal spatial mesh (N m)



Standard solid Fine angular dashed

Low-current power / speed and field torque (N m); gate uses cycle means



Field torque solid Power / speed dashed

Gate	Outcome
mean_torque_delta_pct	PASS
torque_waveform_nrmse_pct	PASS
commutation_transition_delta_samples	PASS
power_torque_delta_pct	PASS
native_standard_to_fine_mean_torque_delta_pct	PASS
native_standard_to_fine_ripple_delta_percentage_points	PASS

Gate	Outcome
native_standard_current_properties	PASS
native_fine_current_properties	PASS
femm_standard_current_properties	PASS
analytical_low_current_current_properties	PASS
native-standard_golden_current_sequence	PASS
native-fine_golden_current_sequence	PASS
analytical_golden_current_sequence	PASS
femm-standard_golden_current_sequence	PASS